

The biregular case of a spectral-inclusion conjecture: the weighted plane class and its obstructions

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Abstract

The conjecture that every root of the d -matching polynomial $\mu_{d,G}$ lies in $\text{spec}(T)$, T the universal cover, is false in general [34]. Its biregular case survives, contains Problem 1 of Song, Fan and Miao, and is the subject of this paper. At $b = 2$ the object is the *weighted plane class*: families $\{(c_k, V_k)\}$ of weighted 2-planes with $\sum_k c_k P_{V_k} \preceq aI$, whose matching polynomial is the mixed characteristic polynomial of Marcus, Spielman and Srivastava.

We prove a positive margin for the complete bipartite family, so no $K_{d,q}$ refutes the repaired statement however wide its gap is made, and we show the margin decays like $n^{-2/3}$, the soft-edge exponent, so the statement is true but tight and no size-free bound can prove it. We give an exact vertex recursion for the plane class, decompose its cross-term into pieces whose nonnegativity would give the band, and prove two unconditional bounds from real-rootedness alone.

The obstructions are the substance. A rank-blind barrier reads the Marchenko–Pastur law and cannot reach the tree band at either edge; compressions cannot reach the inner end at all; the coefficient ladder’s reach is unbounded in the moment order, but an input at a uniform rate is the conclusion it would prove, so the dimension restriction is intrinsic; and the tree moment bound, the natural a priori input, is false off the coordinate case. A ratio route escapes these and settles the inner edge on a nontrivial class. Everything asserted is machine-checked in Lean 4/Mathlib unless labelled otherwise.

Status conventions. Claims here are of three kinds, and the text says which each time rather than leaving it to the reader. A result described as *formalized*, or cited by a Lean declaration name, is machine-checked in Lean 4/Mathlib: the sources are in `RamaLean/` of the accompanying repository, carry no `sorry`, and rest only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A claim described as *measured*, *verified to* a stated tolerance, or *checked numerically* is computational evidence and is not a proof; the script producing it is named, and every named script is run and its output compared against a recorded baseline (`code/cite-check.py`). A claim called a *conjecture* is unproved, and is called that even where the evidence is strong.

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1 Preliminaries, from the companion paper

This paper is a sequel and does not repeat what it rests on. The conjecture, the closed form at first Betti number one, the subdivision theorem, the complete bipartite case and the minimum-degree repair are proved in [34]; they are restated here without proof, in the form used below. Numbering is local to this paper.

Conjecture 1. *For every finite graph G and every $d \geq 1$, all roots of $\mu_{d,G}$ lie in $\text{spec}(T)$, the spectrum of the universal cover of G .*

Conjecture 2 (D3). *Every finite graph of minimum degree at least three satisfies $\text{Zeros}(\mu_G) \subseteq \text{spec}(T_G)$.*

Theorem 3. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) = U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) = \underbrace{(2T_n(x/2) - 2)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}(T_n(x/2))}_{\Psi_r(x)}.$$

Corollary 4.
$$\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = \frac{1}{1 - 2T_n(x/2)z + z^2}, \quad \text{and} \quad \sum_{r \geq 0} \Phi_{n,r}(x) z^r = \frac{(1-z)^2}{1 - 2T_n(x/2)z + z^2}.$$

Corollary 5 (real-rootedness). *All roots of $\Phi_{n,r}$ lie in $[-2, 2]$: with $x = 2 \cos \theta$, χ_{C_n} vanishes at $\theta = 2\pi m/n$ and $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta = m\pi/(rn)$ with $r \nmid m$ (when $r \mid m$ the denominator vanishes too and $\Psi_r = \pm r \neq 0$); this leaves exactly $n(r-1)$ roots, matching $\deg \Psi_r$.*

Corollary 6 (Fibonacci–Lucas evaluation). $\Phi_{n,r}(3) = (L_{2n} - 2) F_{2nr}/F_{2n}$, since $T_n(3/2) = L_{2n}/2$ and $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$.

Corollary 7 (parity). $\Psi_r(-x) = (-1)^{n(r-1)} \Psi_r(x)$: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite C_n it is an even polynomial for every r ; at $r = 2$ it is the classical matching polynomial $2T_n(x/2)$ of C_n , consistent with Godsil–Gutman [6].)

Theorem 8. *Let H be D -regular, $D \geq 2$, and let $S(H)$ be its subdivision, so that the universal cover of $S(H)$ is the $(D, 2)$ -biregular tree with spectral gap $(0, \sqrt{D-1} - 1)$. Then every nonzero root of $\mu_{S(H)}$ has absolute value at least $\sqrt{D-1} - 1$; that is, Conjecture 1 holds for $S(H)$ at $d = 1$. The constant is the spectral gap of the universal cover; we do not claim it is attained, and by the strictness of Heilmann–Lieb on finite graphs it is not.*

Theorem 9. *Let $2 \leq p \leq q$. Every root of $\mu_{K_{p,q}}$ lies in*

$$\{0\} \cup \pm[\sqrt{q-1} - \sqrt{p-1}, \sqrt{q-1} + \sqrt{p-1}],$$

the spectrum of the (q, p) -biregular tree. So Conjecture 1 holds for every $K_{p,q}$ at $d = 1$.

Five displays are used by number below and are reproduced here. The averaging identity

$$\Phi_{n,r}(x) = \mathbb{E}_{\sigma \in S_r} \prod_{\ell \in \text{cycles}(\sigma)} (2T_{n\ell}(x/2) - 2). \quad (1)$$

the recurrence carrying the closed form

$$V_0 = 1, \quad V_1 = \mu_G, \quad V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}. \quad (2)$$

Newton's identity in the form used for the coefficients

$$r \Phi_r = \sum_{k=1}^r f(k) \Phi_{r-k}, \quad \Phi_0 = 1, \quad (3)$$

the arithmetic-geometric mean step

$$\sqrt{u(u+\alpha)} + 1 \leq \sqrt{(u+1)(u+1+\alpha)} \quad (4)$$

and the two-sided statement

$$\text{every root of } g \text{ lies in } [-2\sqrt{m}, 2\sqrt{m}], \quad (5)$$

That is everything borrowed. From here the paper is self-contained, and it proceeds in three movements: what the biregular case has going for it, what the plane class looks like when the question is posed operator-theoretically, and what the obstructions are.

2 Why the biregular case survives, and how far

The refutation leaves the biregular case standing, and this section says why and measures how much room it has: a positive margin for the complete bipartite family, the decay of that margin, the two obstructions that bar compressions and rank-blind barriers from the inner end, and the ratio route that escapes them.

2.1 Why the biregular case survives

For an (a, b) -biregular graph the universal cover is the (a, b) -biregular tree, whose spectrum is $\{0\} \cup \{x : \tau \leq |x| \leq \rho\}$ with $\tau = |\sqrt{a-1} - \sqrt{b-1}|$ and $\rho = \sqrt{a-1} + \sqrt{b-1}$. Since [7] already places every root in $[-\rho, \rho]$, Conjecture 1 for such a graph is *equivalent* to the assertion that no nonzero root satisfies $|x| < \tau$, which is Problem 1 of Song, Fan and Miao. That reduction is machine-checked as `BiregularBlocking.biregular_reduction`, with the tree spectrum and the Hall–Puder–Sawin interval as the two hypotheses; neither is in Mathlib. So the biregular case is not one fragment of the conjecture among others, it is the whole of what survives.

Both mechanisms of [34] are structurally blocked there, and for different reasons. The subcubic construction hangs branches on a tree skeleton, and a graph in which every vertex has degree at least two is not a tree, by handshaking against the edge count of a tree (`no_tree_of_two_le_degree`); a biregular graph with $a, b \geq 2$ has minimum degree at least two and so admits no such skeleton. Hall's construction needs a wide spread of degrees, and

a biregular graph has exactly two. Beyond that, the two constructions place their roots in *bulk* gaps, at 1.20, 2.24 and 2.85 in the examples above, whereas the only gap available in the biregular case is the one at the origin: a counterexample there would need a root just above zero, which is a question about near-deficiency of matchings rather than about band structure.

There is also a quantitative obstruction, in the direction one would want, though it is not complete. At a cut vertex with p isomorphic branches the matching polynomial factors through $F = x\mu_H - p\mu_{H-v}$, and $F(0) = -p\mu_{H-v}(0)$, so a Lipschitz bound gives

$$|\theta| \geq \frac{p|\mu_{H-v}(0)|}{M}, \quad M = \sup |F'|,$$

for every root θ (`branch_root_sep`). The separation grows linearly in the number of branches: strengthening the mechanism drives its roots away from zero, which is the opposite of what a biregular counterexample requires. The bound is conditional on $\mu_{H-v}(0) \neq 0$, that is on $H-v$ having a perfect matching, and Hall's own branch fails that, since there $H-v$ is a star of odd order. It therefore blocks the subcubic mechanism and not his, and on the subcubic branch it gives $|\theta| \geq 1/6$ against a true smallest root of 0.662: valid, and far from sharp.

The vacuity is removable. If $F(0) = 0$, divide out the zero: writing $F = x^m G$ with $G(0) \neq 0$, the nonzero roots of F are the roots of G , and G is what should be bounded. For Hall's branch $F = x^5(x^4 - 16x^2 + 55)$, so $G(0) = 55$ and the obstruction disappears; the bound then gives 0.991 against a true root of $\sqrt{5}$. Carried out on the matching polynomial itself, where $\mu_G(x) = \pm x^{n-2\nu}g(x^2)$ with ν the matching number and $g(0) = \pm m_\nu$, the classical Cauchy estimate becomes a statement purely about matching numbers.

Proposition 10. *Let G be a finite graph with matching number ν and k -matching numbers m_k . Then every nonzero root θ of μ_G satisfies*

$$\theta^2 \geq \frac{m_\nu}{m_\nu + \max_{k < \nu} m_k}.$$

The estimate is Cauchy's and is classical; only its reading in terms of matching numbers is being recorded here. It is machine-checked as `RootSeparation.cauchy_lower` and `matching_root_sep`, with no hypothesis on the graph.

What makes it worth stating is that the inequality points the useful way. By `biregular_reduction`, Conjecture 1 for an (a, b) -biregular graph is exactly the absence of a nonzero root below τ , so any graph whose bound reaches τ satisfies the conjecture, unconditionally and with no appeal to [31]; that implication is `RootSeparation.sfm_of_separation`. Run on random biregular graphs (`code/rootsep.py`) it settles three of nine, the smaller ones, the bound falling as 0.63, 0.45, 0.28 against a fixed $\tau = 0.318$ for $(3, 4)$ -biregular as the graph grows. So it decides small cases outright and is too weak for large ones.

The weakness is not intrinsic: Cauchy's estimate uses only the moduli of the coefficients and discards the one structural fact available. By Heilmann–Lieb μ_G is real-rooted, so g has all roots real and *positive*. For a finite family of positive reals the ℓ^m norm of the reciprocals decreases to the ℓ^∞ norm, so with $P_m = \sum_i y_i^{-m}$,

$$\min_i y_i \geq P_m^{-1/m},$$

sharpening as m grows and exact in the limit. The case $m = 1$ is the harmonic bound $m_\nu/m_{\nu-1}$, already better than Cauchy; and the P_m are computed exactly from the matching numbers by Newton's identities on the reversed polynomial, so every quantity is rational.

Proposition 11. *Let G be biregular with $\tau = |\sqrt{a-1} - \sqrt{b-1}|$, and let $q \geq \tau^2$ be rational. If $P_m q^m \leq 1$ for some $m \geq 1$, then no nonzero root of μ_G has $|\theta| < \tau$, so Conjecture 1 holds for G .*

Proof. $y_i^{-m} \leq P_m \leq q^{-m}$ for each i , hence $y_i \geq q \geq \tau^2$. □

Machine-checked as `PowerSumCertificate.roots_ge_of_powersum`, with division and m -th roots avoided so that the check is exact rational arithmetic. Measured in `code/powersum.py`, $m = 8$ already agrees with the true smallest root to three or four decimal places, and the certificate settles all nine biregular graphs tested, including every size at which the Cauchy and harmonic forms fail.

Run at scale it certifies 417 of 417 random biregular graphs across nine degree pairs and sizes up to 40 vertices, in exact rational arithmetic throughout (`code/certify_biregular.py`). Each of those is an unconditional proof of Conjecture 1 for that graph.

It is a decision procedure and not a proof of the conjecture, and the distinction is not a matter of effort. For a finite family of positive reals the ℓ^m norm of the reciprocals *decreases* to the ℓ^∞ norm, so

$$\sup_m P_m^{-1/m} = y_1$$

exactly: optimising the certificate over m computes the smallest root rather than bounding it. Numerically the agreement is already to five decimal places by $m = 16$ (`code/powersum_limit.py`). Consequently the certificate succeeds for a graph precisely when $y_1 > \tau^2$, which is Song–Fan–Miao for that graph, and a bound on P_m uniform in the size of the graph strong enough to give the conjecture would already be the conjecture. The difficulty has not moved, and the route is recorded as closed rather than promising. What the certificate does supply is a complete and exact decision procedure, which is worth having and is all it is.

Empirically the statement is robust. Over 180 random (a, b) -biregular bipartite graphs on up to 40 vertices, for (a, b) ranging over eight pairs, no root falls below τ ; the smallest nonzero root stays a factor 1.25 to 2.2 above it (`code/biregular.py`). Random sampling is the wrong instrument for a structured phenomenon, so an adversarial local search was run as well, descending on the smallest nonzero root by degree-preserving edge swaps, which stay inside the biregular class exactly. It barely moves the quantity, from 0.6076 to 0.6005 over 1200 swaps in the hardest case, and the final ratios to τ run from 1.42 to 1.97 (`code/biregular_anneal.py`). The smallest nonzero root appears to be rigid inside the biregular class.

That, together with the first Betti number one case where the conjecture is a theorem, is where the remaining interest lies.

2.2 A positive margin for the complete bipartite family

The biregular case is most exposed at large degree contrast, since the gap $(0, \tau)$ widens without bound with q . For $K_{d,q}$ the question is decidable in closed form. A k -matching picks

k vertices on each side and matches them, so

$$\mu_{K_{d,q}} = x^{q-d} P(x^2), \quad P(y) = \sum_{k \leq d} (-1)^k \binom{d}{k} (q)_k y^{d-k}, \quad (6)$$

with $(q)_k$ the falling factorial. Here P is a Laguerre polynomial, and as $q \rightarrow \infty$ its zeros satisfy $(y - q)/\sqrt{q} \rightarrow$ the zeros of the probabilists' Hermite polynomial He_d . Writing h_d for the largest of those, $x_{\min} = \sqrt{q} - \frac{h_d}{2} + o(1)$ against $\tau = \sqrt{q} - \sqrt{d-1} + o(1)$, so the margin tends to $\sqrt{d-1} - h_d/2$; at $q = 10^6$ this agrees to four decimals for every d from 3 to 20 (`code/d3attack.py`).

Theorem 12. *For every $d \geq 3$ the limiting margin is positive: $\sqrt{d-1} - h_d/2 \geq \frac{1}{2}(\sqrt{d-1} - \sqrt{d-2}) > 0$.*

Proof. By the recurrence $\text{He}_{n+1} = y \text{He}_n - n \text{He}_{n-1}$, the zeros of He_d are the eigenvalues of the Jacobi matrix with zero diagonal and off-diagonal entries $\sqrt{1}, \dots, \sqrt{d-1}$. Its absolute row sums are $\sqrt{i-1} + \sqrt{i}$, largest in the last row, so Gershgorin gives $h_d \leq \sqrt{d-2} + \sqrt{d-1}$. $\square$

The Gershgorin step is `CompleteBipartiteMargin.abs_eigenvalue_le_max_rowsum`, proved from the eigenvalue equation at an index where the eigenvector is largest; the consequence is `margin_pos_of_rowsum`, and `no_root_below_gap_edge` closes the chain. No complete bipartite graph violates Conjecture 1, however wide the gap is made.

2.3 The margin vanishes, so the statement is tight

$K_{d,q}$ is not extremal. Random (d, q) -biregular graphs have smaller margins, and the margin falls as they grow: for $(3, 6)$ the minimum over twelve samples falls monotonically across fourteen sizes, from 0.5301 at $n = 12$ to 0.2009 at $n = 51$, fitting a power law at $R^2 \geq 0.999$ in each of twelve families spanning $d = 3$ to 6 (`code/softedge2.py`). Matchings are counted exactly, by a bitmask permanent over the smaller side, which is what makes the range reachable.

The rate is the soft-edge one. The cavity resolvents on the biregular tree satisfy $X = 1/(z - A_1 Y)$, $Y = 1/(z - B_1 X)$ with $A_1 = d - 1$, $B_1 = q - 1$, so $P = XY$ solves $A_1 B_1 P^2 + (A_1 + B_1 - z^2)P + 1 = 0$; writing $a = \sqrt{A_1}$, $b = \sqrt{B_1}$, its discriminant factors,

$$(z^2 - a^2 - b^2)^2 - 4a^2 b^2 = (z^2 - (a + b)^2)(z^2 - (b - a)^2), \quad (7)$$

locating the band edges at $b \pm a$ (`SoftEdge.discriminant_factors`). The inner factor has a simple zero there (`inner_edge_simple`), so the density vanishes like $\sqrt{x - \tau}$ and the density exponent is $\frac{1}{2}$ for every (d, q) . The expected root count within δ of the edge is then $CN\delta^{3/2}$, and the extreme root sits where that count is one, giving $\delta = (CN)^{-2/3}$ (`edge_scale`, `sqrt_edge_exponent`). Writing $\rho(x) = \kappa\sqrt{x - \tau} + O(x - \tau)$, the asymptotic margin is explicit,

$$\delta = \left(\frac{3}{2\kappa n} \right)^{2/3} \quad (8)$$

(`sqrt_edge_quantile`), and agrees with the analytic quantile to four significant figures at $n = 10^6$ in all twelve families. The density itself passes a sharp check: $n \int_{\tau}^{\rho} \rho = r$ to eight decimals, r being the exact number of positive roots.

The consequences are structural, and they disqualify a class of proofs including Theorem 12. A quantity positive at every size may admit no uniform positive lower bound, so any argument producing a constant independent of n is insufficient in general (`no_uniform_lower_bound`); the Gershgorin constant is such a constant. A lower bound may not decay more slowly than the true one, so a proof must reach the exponent (`exponent_floor`). And the picture is a Friedman-type one: the edge is the greatest lower bound of the smallest positive roots and is never attained (`inf_not_attained`), with no constant above it bounding them below (`edge_unimprovable`). There is no ε to spare, and the statement cannot be softened.

Two limits on this. The fitted exponents vary with the aspect ratio q/d at the sizes reachable, from -0.67 at $q = 2d$ to -0.61 at $q = 4d$, and the exponent of the analytic quantile also varies, being $-2/3$ by construction and reaching it only near $n = 10^6$. A finite-size effect is the natural reading, but the check does not establish it: the analytic prediction neither reproduces the measured dependence nor is flat against it, the exponent gap running to 0.093 , so the script returns `INCONCLUSIVE` and the reading stays a reading (`code/quantile.py`). And the observed smallest roots sit where the tree measure places about 0.5 roots, a count that drifts as $n^{+0.10}$; that departure is neither the girth, which is constant at 4 throughout, nor the estimator, since the unbiased sample mean drifts equally, and it is unexplained (`code/nobs.py`).

2.4 Compression cannot reach the inner end

There is a structural reason the outer end of (5) is a theorem and the inner end is not. The outer bound is a statement about quadratic forms: every Rayleigh quotient of a diagonal form lies between its least and greatest entry (`SoftEdge.rayleigh_between`), and being a statement about forms it passes to every subspace. That is exactly the path-tree argument, the path tree being an induced subtree of T and its adjacency form a compression.

A gap is not a statement about quadratic forms and is not inherited. With entries -1 and 1 the spectrum is $\{-1, 1\}$ and $(-1, 1)$ is a gap, yet the vector $(1, 1)$ has Rayleigh quotient 0 , strictly inside it (`rayleigh_in_gap`). So a compression can take values the original excludes, and no argument phrased in Rayleigh quotients or compressions can rule them out however refined. This is the P_8 -inside-the- $(3, 2)$ -biregular-tree phenomenon of [34] at its smallest, and it is why Heilmann–Lieb, Godsil’s path tree and the Marcus–Spielman–Srivastava interlacing machinery all deliver one end of the interval and are silent about the other.

2.5 The ratio route

What is not barred is the ratio recursion, which bounds ratios of matching polynomials rather than Rayleigh quotients. By Godsil μ_G divides μ_P for the path tree P , and on a tree the deletion recurrence is exact and local, $F_v = \lambda - \sum_u 1/F_u$ over the children u of v with $F = \lambda$ at leaves, so $\mu_G(\lambda) \neq 0$ follows once no F vanishes.

A positive invariant interval cannot do it: invariance would need $a \leq \lambda - k/a$, which has a positive solution only for $\lambda \geq 2\sqrt{k}$, the Heilmann–Lieb bound. The certificate is instead sign-alternating. On path trees of (d, q) -biregular graphs at gap points, over twenty-two thousand path-tree vertices, the signs separate by vertex type with no exceptions: every degree- d vertex has $F > 0$ and every degree- q vertex has $F < 0$ (`code/pathratio.py`).

Proposition 13. *Let $\lambda > 0$.*

- (i) *If $F_1, \dots, F_k \in [-C, -c]$ with $c > 0$ then $\lambda \leq \lambda - \sum_j 1/F_j \leq \lambda + k/c$.*
- (ii) *If $F_1, \dots, F_k \in [a, B]$ with $0 < a \leq B$ and $k > \lambda B$, or if $0 < F_j < k/\lambda$ for all j , then $\lambda - \sum_j 1/F_j < 0$.*

These are `RatioRoute.left_step`, `right_step` and `right_step_sharp`; (i) holds for every k , so a path-tree vertex missing some of its children is still covered, and the sharp form of (ii) is what the path trees satisfy, with slack 0.42 to 4.47 and no violations over every vertex measured (`code/childbound.py`).

The combinatorial input is an alternation count. Let π end at $w \in R$, put $k = |N(w) \setminus \pi|$, $p_L = |\pi \cap L|$, $p_R = |\pi \cap R|$, and let u be a child of w with child count k_u .

Lemma 14. *$q - r \leq k$ and $k_u \leq k - (q - r)$.*

Proof. The $q - k$ blocked neighbours of w lie in L , so $q - k \leq p_L$; the path alternates and ends in R , so $p_L \leq p_R \leq r$, giving $q - r \leq k$. For the child, $N(u) \subseteq R$ and an unblocked neighbour must avoid $\pi \cap R$, so $k_u \leq r - p_R \leq r - q + k$. $\square$

The three counting steps are formalized in `PathCount`: `blocked_le_left` and `unblocked_le` are the set-counting steps, and `alt_count` is the alternation count, proved on lists recording which side each path vertex lies on, with `alt_count_last` transferring it from the first vertex to the last. At the minimum $k = q - r$ the lemma forces $k_u = 0$, so the children are leaves of ratio exactly λ and the parent is $\lambda - k/\lambda < 0$ once $k > \lambda^2$.

Assembling both sides gives a system in one constant $m > 0$, the lower bound on $|F|$ at right-type vertices. With $\kappa(t) = \lambda + t/m$ bounding left-type ratios at child count t , and Lemma 14 capping a right-type vertex's children at $k - \Delta$ each where $\Delta = q - r$, the parent is at most $\lambda - k/\kappa(k - \Delta)$ (`RatioRoute.right_step_bound`); the invariant propagates when that is at most $-m$, which is

$$\lambda + m \leq \frac{k}{\kappa(k - \Delta)} \quad \text{for all } \Delta \leq k \leq q - 1 \quad (9)$$

(`closure_iff`, `induction_step`). Equation (9) is a fixed-point condition on m with no closed form; iterating $m \mapsto \min_k (k/(\lambda + (k - \Delta)/m) - \lambda)$ from above gives the largest admissible m .

Theorem 15. *Let G be (d, q) -biregular with r vertices of degree q , and $\lambda \in (0, \tau)$. If (9) admits some $m > 0$, then $\mu_G(\lambda) \neq 0$.*

The recursion consuming the two steps is `PathTreeInduction.invariant`, by strong induction on a height function, with `no_vanishing` concluding that no ratio vanishes; the path tree enters only through an abstraction carrying a side map, a children map and a height that strictly decreases along edges. A single constant is more than the argument needs: the alternation count applies at every depth, a path of ℓ vertices ending in L having $\lfloor \ell/2 \rfloor$ vertices in R , which caps a left-type vertex's children by $\min(d - 1, r - \lfloor \ell/2 \rfloor)$ and bounds a right-type vertex's below by $q - \lfloor \ell/2 \rfloor$. Indexing κ and m by depth turns (9) into a finite backward recursion, and Theorem 15 holds verbatim in that form (`invariant_depth`).

Solving numerically, the condition holds throughout the gap for (d, q, r) equal to $(3, 6, 4)$, $(3, 9, 4)$, $(3, 12, 4)$, $(4, 8, 4)$, $(4, 12, 4)$ and $(5, 10, 4)$, and to 69% of the gap for $(3, 6, 5)$ (`code/depthind.py`); [33] removes that last restriction by bounding each child at its own count rather than at the worst one. This is the first proof of the inner end of (5) for a nontrivial class.

This has to be reconciled with [33], which rules out cavity recursions. The obstruction there is that every b -regular bipartite graph is an induced subgraph of some (a, b) -biregular graph, and over b -regular bipartite graphs the least root tends to zero, so on the closure of the biregular graphs under vertex deletion the inequality to be proved is false; an induction carrying the target statement through that closure cannot work. Theorem 15 is an induction over vertex-deleted subgraphs, but it does not carry the target statement through them. It carries the interval bounds, and those are guarded by $\Delta = q - r \geq 1$: when $\Delta \leq 0$ a right-type path-tree vertex may have no children at all, the sign alternation fails at once, and (9) admits no positive m . A b -regular bipartite graph with parts of equal size has $\Delta = b - r < 0$, so the family witnessing the obstruction is exactly what the hypothesis excludes. The condition $\Delta \geq 1$ is therefore not a technical convenience; it is what [33] forces, and it is the reason the route reaches a nontrivial class rather than nothing or everything.

3 The coefficient ladder

This also explains the boundary between what is proved and what is not, and the explanation is exact.

Theorem 16. *Let G be (a, b) -biregular bipartite with parts of sizes $p \leq q$, degree a on the p -side and b on the q -side, and let $g(z) = f_G(z + c)$ with $c = (a - 1) + (b - 1)$. Then*

$$[z^{p-1}] g = p(b - 2).$$

Consequently the roots of g have mean $-(b - 2)$, and g is the matching polynomial of a graph if and only if $b = 2$.

Proof. Counting edges by their smaller side gives $pa = qb$, so $p \leq q$ forces $a \geq b$ and $b = \min(a, b)$. The polynomial f_G is monic of degree p , and its next coefficient is $-m_1$, where m_1 is the number of 1-matchings, that is $|E| = pa$. Hence

$$[z^{p-1}] f_G(z + c) = \binom{p}{1} c - pa = p(a + b - 2) - pa = p(b - 2).$$

The mean of the roots is $-[z^{p-1}]g/p = -(b - 2)$. A matching polynomial of degree p involves only every other power of z , so its coefficient at z^{p-1} vanishes; thus $b = 2$ is necessary. It is sufficient: for $b = 2$ the graph is $S(H)$ for an a -regular H and $g = \mu_H$ by [11]. $\square$

So for $b = 2$ the polynomial g is a matching polynomial, Heilmann–Lieb applies verbatim, and one recovers Theorem 8 and the subdivision identity as the case $c = (a - 1) + 1$ of the change of variable. For $b \geq 3$ it provably is not, and Theorem 16 says more than that it fails: it locates the failure. The roots of g are not centred at the band centre but sit a distance $b - 2$ below it, towards the lower band edge, which is exactly the end of (5) that is open.

The asymmetry of the problem and the departure from the matching-polynomial class are the same phenomenon, measured by the single quantity $b - 2$. Examples:

$$g_{K_{3,4}} = z^3 + 3z^2 - 9z - 19, \quad g_{K_{3,5}} = z^3 + 3z^2 - 12z - 24, \quad g_{K_{4,5}} = z^4 + 8z^3 - 6z^2 - 128z - 139,$$

with subleading coefficients $3 = 3(3 - 2)$, $3 = 3(3 - 2)$ and $8 = 4(4 - 2)$ as the theorem requires.

The same method determines more than the one coefficient, and shows that the graph itself does not enter until surprisingly late.

Theorem 17. *With notation as in Theorem 16, the coefficients of z^p , z^{p-1} , z^{p-2} and z^{p-3} in g depend only on (p, a, b) , not on G . Explicitly the first three are 1, $p(b - 2)$ and $\frac{p}{2}[(p - 1)(b - 2)^2 - a(b - 1)]$.*

Proof. The coefficients of g up to z^{p-j} are determined by $m_1, \dots, m_j$, so it suffices that m_1, m_2, m_3 are functions of (p, a, b) alone. Write $N = |E| = pa = qb$. Then $m_1 = N$. Two distinct edges of a bipartite graph meet in at most one vertex, so $m_2 = \binom{N}{2} - p\binom{a}{2} - q\binom{b}{2}$, which after substituting $q = pa/b$ gives $m_2 = \frac{pa}{2}(pa - a - b + 1)$; the stated value of $[z^{p-2}]g$ follows from $\binom{p}{2}c^2 - (p - 1)Nc + m_2$ with $c = a + b - 2$.

For m_3 , let L be the graph on E in which two edges are adjacent when they meet, so m_3 is the number of independent 3-sets of L . Every vertex of L has degree $(a - 1) + (b - 1) = c$. Three pairwise meeting edges of a bipartite graph must share a vertex, since otherwise they form a triangle in G ; hence the number of triangles of L is $t = p\binom{a}{3} + q\binom{b}{3}$, and the number of 2-edge paths is $N\binom{c}{2} - 3t$. The standard count $\binom{N}{3} - \frac{Nc}{2}(N - 2) + (N\binom{c}{2} - 3t) + 2t$ therefore depends only on (p, a, b) . $\square$

The graph enters at the very next coefficient, and it does so through a single invariant, with coefficient one.

Theorem 18. *Let $C_4(G)$ denote the number of 4-cycles of G . Then*

$$[z^{p-4}]g = U(p, a, b) + C_4(G),$$

where, writing $\beta = b - 2$ and $A = a(b - 1)$,

$$U = \binom{p}{4}\beta^4 - \frac{3}{2}A\beta^2\binom{p}{3} + \frac{A(3A + 4\beta^2)}{12}\binom{p}{2} - \frac{A(9A + 2b^2 - 14b + 14)}{24}p.$$

Proof. As in Theorem 17, m_4 is the number of independent 4-sets of L , and inclusion-exclusion expresses it through the *non-induced* counts in L of the ten graphs on at most four vertices with no isolated vertex. Nine of the ten are universal, by the clique structure of L : each vertex of L is a cell of the $p \times q$ biadjacency matrix, and two cells are adjacent exactly when they share a row or share a column, never both, so every edge of L has a well-defined type.

Only the count of C_4 in L is not universal. Reading the cyclic word of types around a four-cycle of L , the words $RRRC$ and $RCCC$ are impossible, since three consecutive edges of one type force all four cells into a single line and then the fourth edge has that type too; and $RRCC$ is impossible, since it would require a cell sharing a column with two distinct cells of one row. This leaves $RRRR$ and $CCCC$, whose four cells lie in one line and which

therefore number $3[p\binom{a}{4} + q\binom{b}{4}]$, and $RCRC$, whose four cells are exactly the four entries of a 2×2 all-ones submatrix, that is a 4-cycle of G , each contributing exactly one. Substituting and applying the shift $[z^{p-4}](z+c)^{p-k} = \binom{p-k}{p-4}c^{4-k}$ gives the stated U . $\square$

Together with Theorem 17 this determines the top five coefficients of g completely:

$$\begin{aligned} [z^{p-1}]g &= p\beta, & [z^{p-2}]g &= \binom{p}{2}\beta^2 - \tfrac{1}{2}Ap, \\ [z^{p-3}]g &= \binom{p}{3}\beta^3 - A\beta\binom{p}{2} + \tfrac{1}{6}A\beta p, & [z^{p-4}]g &= U + C_4(G). \end{aligned}$$

For the Heawood graph ($p = 7$, $a = b = 3$, $C_4 = 0$) the last reads $35 - 315 + 231 - 77 = -126$. So the conjecture is insensitive to the graph until girth enters, and it enters at the first opportunity: four pairwise disjoint edges are first constrained by a 4-cycle.

Theorem 17 has a consequence for what a proof of Conjecture 1 can look like. The universal coefficients determine the first two power sums of the roots of g , namely $\sum \rho_i = -p(b-2)$ and $\sum \rho_i^2 = p[(b-2)^2 + a(b-1)]$, and those two numbers already imply a bound on the least root.

Proposition 19. *Every root of g satisfies $\rho \geq -(b-2) - \sqrt{(p-1)a(b-1)}$, and this is the sharpest bound implied by the coefficients of z^p, z^{p-1}, z^{p-2} .*

Proof. The roots are real. Fix a root ρ and apply Cauchy–Schwarz to the other $p-1$: $(\sum \rho_i - \rho)^2 \leq (p-1)(\sum \rho_i^2 - \rho^2)$. With $e = \sum \rho_i$ and $S = \sum \rho_i^2$ this is $p\rho^2 - 2e\rho + e^2 - (p-1)S \leq 0$, so $\rho \geq [e - \sqrt{(p-1)(pS - e^2)}]/p$. Substituting the two power sums gives $pS - e^2 = p^2a(b-1)$ and the stated bound. Sharpness is the equality case of Cauchy–Schwarz. $\square$

The bound of Proposition 19 grows like $\sqrt{p}$, whereas the conjectured bound $-2\sqrt{m}$ does not depend on p at all. So for all but the smallest graphs the moment bound is far weaker, and since by Theorem 17 the coefficients it uses do not see G , no argument resting on them can distinguish one (a, b) -biregular graph from another. Any proof of Conjecture 1 must therefore use m_4 or beyond, that is, it must be sensitive to the cycle structure of G . This is consistent with the place the conjecture occupies: bounds of the shape $2\sqrt{(a-1)(b-1)}$ are non-backtracking bounds, and the non-backtracking operator of a bipartite graph is exactly the object that records cycles. That object can be exhibited, but only in averaged form, and the reason for the averaging is not incidental.

4 The operator form

4.1 The mixed characteristic polynomial

The multivariate object that does carry the biregular information is the mixed characteristic polynomial of Marcus, Spielman and Srivastava [22]: with $P_k = \text{diag}(\mathbf{1}_{N(k)})$ of rank b and $\sum_k P_k = aI_p$, one has $\nu_G = \mu[P_1, \dots, P_q]$, since $\sum_k z_k P_k$ is diagonal and each ∂_{z_k} must strike a distinct linear factor, so the surviving terms are the injections, that is the matchings. This is the stability route done properly, and it locates the shortfall quantitatively: a barrier argument that sees only the rank, the trace and $\sum_k P_k = aI$ is reading the Marchenko–Pastur law, and yields the interval $[(\sqrt{a} - \sqrt{b})^2, (\sqrt{a} + \sqrt{b})^2]$, which is strictly wider than

the band, the discrepancy being $2[\sqrt{ab} - \sqrt{(a-1)(b-1)} - 1] \geq 0$ with equality only at $a = b$. So the standard potentials do not even recover the known upper bound. That is not a deficiency of the analysis but of the hypotheses, and it can be seen exactly. Take $A_k = (b/p)I_p$: these are positive semidefinite with $\text{tr } A_k = b$ and $\sum_k A_k = (qb/p)I = aI$, so they satisfy every hypothesis the barrier method uses, and $\mu[A_1, \dots, A_q]$ is a Laguerre polynomial whose roots converge to the Marchenko–Pastur edges. At $(a, b) = (3, 2)$ the upper edge is already exceeded at $p = 4$, where the largest root is 6.229 against $(s+t)^2 = 5.828$, and the lower edge at $p = 48$, where the smallest is 0.154 against $(s-t)^2 = 0.172$. The Marchenko–Pastur interval is therefore attained on those hypotheses, and *no* barrier argument that does not see $\text{rank } A_k \leq b$ can reach the tree band at either edge. Numerically the rank condition is what carries the statement: dropping idempotency but keeping $\text{rank } A_k \leq b$ and $\text{tr } A_k = b$ produced no violation in search, which suggests that class, rather than the projections, as the natural target. What is missing is a barrier built from the Kesten measure of the (a, b) -biregular tree rather than from Marchenko–Pastur, which is to say one that uses that the P_k are 0/1 diagonal, exactly the combinatorial information free probability discards.

That raises the question of how much of the hypothesis is really needed, and the answer delimits the problem usefully. Asking for the same conclusion from *arbitrary* positive semidefinite A_k of rank b with $\sum_k A_k = aI_p$ is false. Take $p = 2$, $q = 3$, $(a, b) = (3, 2)$, and $A_1 = A_2 = \frac{u}{2}I_2$, $A_3 = (3-u)I_2$: each has rank 2 = b and the sum is $3I = aI$, while $\mu[A_1, A_2, A_3](y) = y^2 - 6y + (6u - \frac{3}{2}u^2)$. At $u = \frac{1}{10}$ the constant term is $\frac{117}{200}$ and the roots are 0.09914 and 5.90086, outside the band $[3 - 2\sqrt{2}, 3 + 2\sqrt{2}] = [0.17157, 5.82843]$ at both ends; as $u \rightarrow 0$ the roots tend to 0 and 6. The counterexample is diagonal, hence commuting, so noncommutativity is not what fails. What fails is that the traces are unconstrained: $\text{tr } A_3 = 5.8 \gg b$. Imposing $\text{tr } A_k = b$ restores the second coefficient, since it then depends on the family only through $\sum_k \|A_k\|_F^2$, which by Cauchy–Schwarz is at least b with equality exactly for idempotents. Projections are thus the extreme point already at the first nontrivial coefficient.

For rank- b orthogonal projections the statement survives, once one adds the hypothesis $a \geq b$, equivalently $p \leq q$, which the graph setting supplies automatically. Without it the statement is false for a trivial reason: if $p > q$ then y^{p-q} divides $\mu[P_1, \dots, P_q]$, so 0 is a root while the lower band edge is positive. Taking $p = 6$, $q = 4$, $(a, b) = (2, 3)$ with $P_1 = P_2$ and $P_3 = P_4$ the projections onto two complementary 3-spaces gives a double root at 0 against a lower edge of $(\sqrt{2} - 1)^2$. A search over more than 10^5 feasible families, real and complex projections, and the intermediate class $\text{rank} \leq b$, $\text{tr} = b$, produced no violation, the smallest observed margin being 3.6% of the band width. That figure needs a caveat, though, and it cuts against the aggregate impression: every tight case in those searches has $b = 2$, hence lies in the regime settled by Proposition 38. In the genuinely open range $b \geq 3$ the observed margins are wide, at $(p, q, a, b) = (6, 8, 4, 3)$ the least root is 0.486 against a band edge of 0.101, a factor of nearly five. The numerical evidence for the open part of the statement is thinner than the total counts suggest. Two features make this the right generalization to record. First, commuting rank- b projections with $\sum_k P_k = aI$ are simultaneously diagonalizable and therefore 0/1 diagonal, so they are *exactly* the biregular incidence families: the projection statement is precisely the noncommutative extension of Conjecture 1. Second, the universality of Theorem 17 persists in that setting, for projections the coefficients $E_0, \dots, E_3$ depend only on (p, q, a, b) , because idempotency collapses every trace of a word of length at most three with

a repeated letter, while at length four $\text{tr}(A_i A_j A_i A_k)$ does not reduce. The first free invariant sits at level four on both sides of the analogy, matching the 4-cycle count of Theorem 18.

4.2 A determinantal representation

The mixed characteristic polynomial admits a probabilistic representation which gives an unconditional bound with no barrier or stability input. Write $P_k = W_k W_k^*$ with $W_k^* W_k = I_b$, set $U = [W_1 | \cdots | W_q] / \sqrt{a}$, and let $\Pi = U^* U$, a rank- p orthogonal projection on $\mathbb{R}^n$, $n = pa = qb$, whose q diagonal $b \times b$ blocks are all $\frac{1}{a} I_b$. Let S be a sample from the determinantal process with kernel Π and $s_k(S)$ the number of slots of block k it uses. Then

$$y^{q-p} \mu[P_1, \dots, P_q](y) = \mathbb{E}_S \prod_{k=1}^q (y - a s_k(S)), \quad (10)$$

by expanding $\det(yI + \sum_k z_k P_k)$ over subsets and applying $\prod_k (1 - \partial_{z_k})$, which annihilates every term using a block twice. Here $|S| = p$ always, so $\sum_k s_k = p$, and each s_k is exactly $\text{Binomial}(b, 1/a)$; for a diagonal family S is simply an independent uniform choice of one incident edge at each vertex of the p -side.

Since $0 \leq s_k \leq b$, every factor in (10) is positive once $y > ab$, so the expectation is, and there is no root there.

Proposition 20. *Every root of $\mu[P_1, \dots, P_q]$ is at most ab . Moreover, with $m = (a-1)(b-1)$,*

$$ab - (\sqrt{a-1} + \sqrt{b-1})^2 = (\sqrt{m} - 1)^2.$$

So the bound overshoots the conjectured edge by exactly $(\sqrt{m} - 1)^2$, and is lossless precisely when $m = 1$. At $a = b = 2$ the overshoot vanishes and $ab = 4$ is the upper edge; the roots are nonnegative because the matching numbers are, so **the band is proved outright at $a = b = 2$.**

Where ab stands relative to what is known needs care, and we record it exactly. Xu, Xu and Zhu [5, Thm. 1.9] prove, for positive semidefinite $X_1, \dots, X_q$ of rank at most k with $\sum X_i \preceq I$ and $\varepsilon = \max \text{tr } X_i$, that $\text{maxroot } \mu \leq (\sqrt{1 - \varepsilon/(k-1)} + \sqrt{\varepsilon})^2$ whenever $\varepsilon \leq (k-1)^2/k$. Normalizing a rank- b projection family with $\sum_k P_k = aI$ by a puts it in exactly that form with $k = b$ and $\varepsilon = b/a$, and since the roots scale with the family this gives

$$\lambda_{\max} \leq a \left(\sqrt{1 - \frac{b}{a(b-1)}} + \sqrt{\frac{b}{a}} \right)^2, \quad a \geq \frac{b^2}{(b-1)^2}, \quad (11)$$

which at $b = 2$ is $a + 2\sqrt{2a-4}$ for $a \geq 4$. That is strictly better than ab for every $a > 4$, indeed $a + 2\sqrt{2a-4} \leq 2a$ with equality only at $a = 4$, formalized as `xxz_le_ab` and `xxz_eq_ab_iff`, and better than Marchenko–Pastur throughout. So ab is the best available bound only where the hypothesis of (11) fails, which at $b = 2$ means $a < 4$: at $a = b = 2$, where it is sharp and settles the band, and at $(a, b) = (3, 2)$, where it gives 6 against a Marchenko–Pastur edge of

9.899 and a band edge of 5.828. Elsewhere (11) is the state of the art:

(a, b)	(11)	ab	Marchenko–Pastur	$a + 2\sqrt{a}$	band
$(3, 2)$	n/a	6.000	9.899	6.464	5.828
$(4, 2)$	8.000	8.000	11.657	8.000	7.464
$(9, 2)$	16.483	18.000	19.485	15.000	14.657
$(100, 2)$	128.000	200.000	130.284	120.000	119.900
$(4, 3)$	10.977	12.000	13.928	—	9.899
$(25, 4)$	47.126	100.000	49.000	—	43.971

The comparison also says what the target of this section is worth: at $b = 2$ the bound $a + 2\sqrt{a}$ of Proposition 24 would improve on (11) for every $a > 4$, since $\sqrt{a} \leq \sqrt{2a - 4}$ there (`target_le_xxz`). It is therefore not a weaker target than the literature already provides, but a stronger one everywhere the barrier hypothesis holds. Proposition 20 is formalized in `RamaLean/ProductBound.lean` as `no_root_above` and `edge_gap`.

The representation also supports an exact recursion on the underlying kernels. Writing N_K for the transversal expansion attached to a kernel K , one has the border identity

$$a N_{M+ww^\top} = (a - y) N_M + N_{\begin{pmatrix} 0 & w^\top \\ w & M \end{pmatrix}},$$

and Schur complements of admissible kernels remain admissible, with $\delta_e \leq 1 - c_e$ forced by $K^2 \preceq K$. This makes an induction on kernel size available in principle. It does not close the upper edge: the scalar cavity invariants that carry the classical argument on graphs are false on the kernel class, witnessed by explicit rational kernels on which the graph unfolding identity fails as an inequality by a y -independent amount. What survives is a reduction of the upper edge on the whole class to a single inequality on bordered kernels, proved for at most two blocks at $b = 2$ by monotonicity of transversal determinants under downdates, which follows from positivity of the adjugate of a positive semidefinite matrix.

4.3 The case of rank two

At $b = 2$ the representation (10) collapses further. Only $s_k \in \{0, 1, 2\}$ occurs, and $\sum_k s_k = p$ forces the counts of 0s and 1s to be determined by $n_2 = \#\{k : s_k = 2\}$, so with $\Psi(u) = \mathbb{E} u^{n_2}$ one has

$$\mu(y) = (y - a)^p \Psi(\theta(y)), \quad \theta(y) = 1 - \frac{a^2}{(y - a)^2}.$$

Real-rootedness of μ forces Ψ to be real-rooted with nonpositive roots, hence n_2 a sum of independent Bernoulli variables, and the extreme roots of μ are $a(1 \pm \sqrt{\pi_{\max}})$ for $\pi_{\max}$ the largest of those parameters. Since the band at $b = 2$ is $[a - 2\sqrt{a - 1}, a + 2\sqrt{a - 1}]$, both edges say $a\sqrt{\pi_{\max}} \leq 2\sqrt{a - 1}$:

Proposition 21. *For $b = 2$, Conjecture 1 for the projection family is equivalent to*

$$\pi_{\max} \leq \frac{4(a - 1)}{a^2} = 1 - \left(\frac{a - 2}{a}\right)^2.$$

The equivalence of the two edges with each other and with this inequality is formalized in `RamaLean/ProductBound.lean` as `band_iff_pi`; that they coincide is Proposition 38 seen through θ , which is two-to-one about $y = a$.

4.4 The matching form

Recentring at a turns μ into a matching polynomial, and the coefficients have a basis-free form. They also have a Grassmann form: writing $A_k = b_{k1}b_{k1}^\top + b_{k2}b_{k2}^\top$ and $\omega_k = b_{k1} \wedge b_{k2} \in \Lambda^2 \mathbb{R}^p$, the Cauchy–Binet formula gives $e_{2r}(\sum_{k \in T} A_k) = \|\bigwedge_{k \in T} \omega_k\|^2$, so

$$\mu(x + a) = \sum_{T \subseteq [q]} (-1)^{|T|} \left\| \bigwedge_{k \in T} \omega_k \right\|^2 x^{p-2|T|},$$

and M_r is a sum of squared Plücker norms. In the commuting case each norm is 0 or 1 and one recovers the matching numbers.

Theorem 22. *Let $P_1, \dots, P_q$ be rank-two orthogonal projections on $\mathbb{R}^p$ with $\sum_k P_k = aI$. Then*

$$\mu(x + a) = \sum_{r \geq 0} (-1)^r M_r x^{p-2r}, \quad M_r = \sum_{|T|=r} e_{2r} \left(\sum_{k \in T} P_k \right),$$

where e_m denotes the m -th elementary symmetric function of the spectrum. The M_r are nonnegative, $M_0 = 1$, $M_1 = q$, and $M_r \leq \binom{q}{r}$.

The expression for M_r involves no dilation and no choice of basis for the ranges of the P_k . When the P_k commute, $\sum_{k \in T} P_k$ is a 0/1 diagonal matrix of rank $2r$ exactly when T is a matching of the multigraph whose edges are the blocks, and has rank below $2r$ otherwise; so e_{2r} is the indicator of that event and M_r is the number of r -matchings. Thus at $b = 2$ the recentred mixed characteristic polynomial is the matching polynomial in the commuting case, and M_r is its natural noncommutative extension. Verified exactly on $S(K_4)$, $S(K_{3,3})$, the cube, and on noncommuting families including the icosahedral one, where the commutators have norm 0.57.

Two consequences change how the remaining gap should be described. First, in the commuting case Proposition 21 is Heilmann–Lieb, so it is a theorem there, and it is sharp: the extremal families are the large-girth a -regular graphs, for which the ratio of $\pi_{\max}$ to its bound rises monotonically through 0.866 at $p = 10$ to 0.949 at $p = 38$. No argument with slack can close it. Second, and more to the point, in that formulation Proposition 21 says exactly that the roots of a mixed characteristic polynomial lie inside the support of the corresponding free convolution, which is the known open sharpening of the Marcus–Spielman–Srivastava bound. So $b = 2$ is not an easier case awaiting a short argument; it is that open problem in another notation.

4.5 The plane class, and two unconditional bounds

This subsection is the longest in the note, so its outcome is stated first. It delivers an exact vertex recursion for the plane class (Theorem 23), checked numerically to 10^{-13} ; a decomposition of its cross-term into pieces C_r (Proposition 24, Theorems 25 and 26) whose nonnegativity would give the band $[a - 2\sqrt{a}, a + 2\sqrt{a}]$ for all weighted plane families, supported numerically but conjectural; and two bounds following unconditionally from real-rootedness alone, giving the band whenever $m \leq 8$ and whenever $m \leq 16 + (\sum_k c_k^2)/a^2$, the latter reproducing the known two-moment bound for projections and extending it to all weighted plane families. The general case is not closed.

What the problem is, at $b = 2$, can be said exactly. Both $\mu(x + a)$ and the operator $\text{Adj}(A) := \sum_k e_2(A_k) P_{\text{range } A_k}$ depend on the family only through the pairs $(e_2(A_k), \text{range } A_k)$. So the class is the class of weighted 2-plane families $\{(c_k, V_k) : c_k \geq 0, \sum_k c_k P_{V_k} \preceq aI\}$, and $\mu(x + a)$ is the matching polynomial of such a family; for coordinate planes it is the ordinary weighted matching polynomial. Two points about that identification need care, and both are invisible in the projection case.

First, the invariance holds within the tight class and not outside it: with $\sum_k A_k$ a multiple of the identity, families sharing the pairs $(e_2(A_k), \text{range } A_k)$ have the same recentred μ even when the A_k are strongly anisotropic, verified to 10^{-13} at eigenvalue ratios up to 129 (`code/rl_conj.py`); without tightness the recentring has no meaning.

Second, the recentring constant is that tight constant, and it is *not* the constant in $\text{Adj}(A) \preceq aI$. Plane data with $\text{Adj}(A) = aI$ admits a realization with $\sum_k A_k = a'I$ — solvable to 10^{-16} — but with $a' > a$ strictly: $a' = 3.860$ against $a = 3$ at $m = 4$, and 4.385 against 3 and 5.120 against 4 at $m = 6$ (`code/rl_conj.py`). So for weighted families $F_A(x) = \mu(x + a')$ with $a' \neq a$, and the identity as stated holds exactly when the two constants agree, which is the projection case $A_k = P_k$, where $e_2 \equiv 1$ gives $\sum_k A_k = \text{Adj}(A) = aI$. Everything below is derived from the wedge coefficients M_r and from $\text{Adj}(A)$ directly and is unaffected; what does not transfer automatically to the weighted class is a result stated for mixed characteristic polynomials, since the two constants there are different.

Consequently the band is stated below on the roots of F_A , as $|x| \leq 2\sqrt{a}$ with a the constant of $\text{Adj}(A) \preceq aI$, and that is the form the unconditional bounds establish, through $\lambda^2 \leq M_1 \leq am/2$. Written in the μ variable it becomes $[a' - 2\sqrt{a}, a' + 2\sqrt{a}]$: the radius is set by the Adj constant and the centre by the tight one. Both the shift $y = x + a'$ and the bound $|x| \leq 2\sqrt{a}$ are checked directly (`code/rl_conj.py`); the two coincide precisely for projection families, where $a' = a$, and every μ -variable statement of the band in this paper should be read with that proviso.

The band being sought is the sharp weighted Heilmann–Lieb constant for plane families, and that constant is not $2\sqrt{a}$. On projections the conjectural sharp value is $2\sqrt{a-1}$, the spectral radius of the a -regular tree, and $2\sqrt{a-1} < 2\sqrt{a}$ strictly (`XuBound.xu_lt_note_target`); it is the case $b = 2$ of the bound $(\sqrt{a-1} + \sqrt{b-1})^2$ on maxroot μ conjectured by Xu¹, which is exactly the upper edge of the (a, b) -biregular tree band. So $2\sqrt{a}$ is the ceiling of the cavity argument used here, not the truth, and an argument of this kind cannot be sharp. That the sharp value is $(\sqrt{a-1} + \sqrt{b-1})^2$ and not something smaller is no longer conjectural on the coordinate locus: it is [33], at every b .

In that language the vertex step has an exact form. Setting $\Theta^{(r)} := \sum_{|T|=r} M_{\omega_T} \succeq 0$ and $W(x) := \sum_{r \geq 1} (-1)^r \Theta^{(r)} x^{m-2r}$, one has $\langle e, W(x)e \rangle = F_A(x) - x F_{A^{(e)}}(x)$ for every unit vector e , where $F_A(x) = \mu(x + a)$ and $A^{(e)}$ is the compression to $e^\perp$; consequently $\text{tr } W = mF_A - xF'_A$ and $F'_A = \sum_i F_{A^{(e_i)}}$ for any orthonormal basis. The cavity hypothesis is then equivalent to a single matrix inequality: when $\text{Adj}(A) = aI$, the band $[a - 2\sqrt{a}, a + 2\sqrt{a}]$ holds if and only if

$$W(2\sqrt{a}) + F_A(2\sqrt{a})I \succeq 0,$$

a coordinate-free statement with no recursion in it.

¹Personal communication, August 2026, deriving it from a generalisation of [30, Conjecture 1].

The matrix W has a closed form. If $\omega_T = c_1 \wedge \cdots \wedge c_{2r}$ is decomposable with factor matrix $C = [c_1 | \cdots | c_{2r}]$ and Gram matrix $G = C^\top C$, then $\langle \omega_{\hat{j}}, \omega_{\hat{l}} \rangle$ is the (l, j) cofactor of G , so

$$M_{\omega_T} = C \operatorname{adj}(G) C^\top, \quad (12)$$

whence $\operatorname{tr} M_{\omega_T} = 2r \|\omega_T\|^2$ and $\operatorname{tr} \Theta^{(r)} = 2r M_r$, which is the identity $\operatorname{tr} W = m F_A - x F'_A$ again. Two of the levels are forced to be scalar: $\Theta^{(1)} = \operatorname{Adj}(A) = aI$ by hypothesis, and when m is even $\Theta^{(m/2)} = M_{m/2} I$, since a top-degree form is a multiple of the volume form and $\|\iota_v \omega\|^2 = \|\omega\|^2 \|v\|^2$ for such. Hence for $m \leq 4$ every level is scalar, $W(x)$ is a scalar matrix, and the matrix inequality degenerates to a scalar one; from $m = 5$ on it does not, and that is where the content begins.

Behind the matrix inequality there is a vertex recursion valid in every direction, which the graph case sees only along the coordinate axes.

Theorem 23. *Let A be a weighted 2-plane family on $\mathbb{R}^m$ with bivectors ω_k , and let e be a unit vector. Put $f_k = \iota_e \omega_k$, so that*

$$\omega_k = e \wedge f_k + \omega'_k, \quad f_k, \omega'_k \in e^\perp,$$

and ω'_k is the bivector of the compressed family $A^{(e)}$. Then for every $r \geq 1$

$$M_r(A) = M_r(A^{(e)}) + \sum_{|T|=r} \left\| \sum_{k \in T} f_k \wedge \omega'_{T \setminus k} \right\|^2,$$

and consequently, with $N_r(e)$ denoting that sum of squares,

$$F_A(x) = x F_{A^{(e)}}(x) - N_e(x), \quad N_e(x) = \sum_{r \geq 1} (-1)^{r-1} N_r(e) x^{m-2r}.$$

Proof. Since e is a unit vector, $\iota_e(\omega_k - e \wedge f_k) = f_k - f_k = 0$, so $\omega_k - e \wedge f_k$ lies in $\Lambda^2(e^\perp)$; writing $b_i = \beta_i e + b'_i$ identifies it as $b'_1 \wedge b'_2$, the bivector of the compressed block. As $e \wedge f_k$ is a 2-form it commutes with the other factors, and $e \wedge e = 0$ kills every term using two of them, so $\omega_T = \omega'_T + e \wedge \sum_{k \in T} f_k \wedge \omega'_{T \setminus k}$. The two summands lie in $\Lambda^{2r}(e^\perp)$ and $e \wedge \Lambda^{2r-1}(e^\perp)$, which are orthogonal, so the norms add; summing over $|T| = r$ gives the first display, and the second is that display multiplied by $(-1)^r x^{m-2r}$ and summed. $\square$

We have looked for Theorem 23 in the literature and not found it. The whole arXiv corpus mentioning “mixed characteristic polynomial” is seven papers, all working through the barrier function or interlacing families; the generalizations of Heilmann–Lieb we can find are to hypergraphs [4], to spanning subgraphs with degree constraints, and to multivariate independence polynomials, all of them graph-theoretic; and a search of arXiv for “matching polynomial” together with any of Grassmann, Plücker, bivector or exterior algebra returns nothing. That is weak evidence: the Gram form below is elementary enough to be folklore, the pre-arXiv matching-polynomial literature is not covered by such a search, and the statement could well appear under different language, hyperbolic or Lorentzian polynomials, or the theory of stable polynomials, where Plücker coordinates do occur but in the different role of characterizing stability by total nonnegativity. We claim only that we have not found it.

The exterior algebra can be dispensed with entirely, and doing so identifies the theorem with a known lemma. Let C be the $m \times 2r$ matrix whose columns are the factors of ω_T , so that $\|\omega_T\|^2 = \det(C^\top C)$. Splitting off e is the decomposition $C = eg^\top + C'$ with $g = C^\top e$ and $C' = (I - ee^\top)C$ the compressed factor matrix; since $e^\top C' = 0$ and $\|e\| = 1$,

$$C^\top C = C'^\top C' + gg^\top.$$

So the content of Theorem 23 is the *matrix determinant lemma*

$$\det(A + gg^\top) = \det A + g^\top \operatorname{adj}(A) g, \quad A = C'^\top C', \quad (13)$$

and the nonnegativity of the cavity term is nothing but positivity of the adjugate of a positive semidefinite matrix. In particular $M_r(A) \geq M_r(A^{(e)})$ needs no computation at all. Splitting g into its r blocks of two identifies the pieces: the block-diagonal part of the quadratic form (13) is $\sum_k \theta_k M''_{k,r-1}$ and the off-block-diagonal part is C_r , which is how the cross terms should be read.

For a coordinate family at a coordinate direction $e = e_v$ one has $f_k = \pm e_u$ for the edges $k = uv$ and $f_k = 0$ otherwise, the cross terms vanish, and $N_r(e_v) = \sum_{u \sim v} m_{r-1}(G - v - u)$: the theorem is then the Heilmann–Lieb vertex recursion. At a generic direction, or for a family that is not diagonal, the cross terms do not vanish and the recursion is not the classical one. Two things follow at once. First, taking r arbitrary, $M_r(A) \geq M_r(A^{(e)})$ for every unit e : the matching numbers of a plane family decrease under compression in every direction, which for graphs is $m_r(G) \geq m_r(G - v)$. Second, the whole band reduces to the sign of the cross terms.

Proposition 24. *Write $X_e := N_e - \sum_k \theta_k F_{A_k''}$ for the cross-term part of N_e , where $\theta_k = \|f_k\|^2$ and A_k'' is the compression to $\{e, f_k\}^\perp$. If $X_e(x) \leq 0$ for every weighted 2-plane family with $\operatorname{Adj} \preceq aI$, every unit e and every $x \geq 2\sqrt{a}$, then every such family satisfies the band $[a - 2\sqrt{a}, a + 2\sqrt{a}]$.*

Proof. Induct on m . Put $R_e = F_A/F_{A^{(e)}}$ and $R'_k = F_{A^{(e)}}/F_{A_k''}$. Dividing Theorem 23 by $F_{A^{(e)}}$,

$$R_e = x - \sum_k \frac{\theta_k}{R'_k} - \frac{X_e}{F_{A^{(e)}}}.$$

The class is closed under compression, so the inductive hypothesis $R'_k \geq x/2$ applies, and $\sum_k \theta_k = \sum_k \|\iota_e \omega_k\|^2 = \langle e, \operatorname{Adj}(A)e \rangle \leq a$. Hence $R_e \geq x - 2a/x$, and $x - 2a/x \geq x/2$ exactly when $x \geq 2\sqrt{a}$. The base case $m \leq 1$ is $R_e = x$. So $R_e \geq x/2 > 0$ for $x \geq 2\sqrt{a}$, and $F_A = R_e F_{A^{(e)}}$ has no root there; F_A is even or odd, so all roots lie in $[-2\sqrt{a}, 2\sqrt{a}]$. $\square$

The leading cross term can be evaluated exactly, and it is a square.

Theorem 25. *Let A be a weighted 2-plane family on $\mathbb{R}^m$ and e a unit vector, and put $f_k = \iota_e \omega_k$, $\omega'_k = \omega_k - e \wedge f_k$ and $u_k = \iota_{f_k} \omega'_k$. Then*

$$C_2 = \left\| \sum_k f_k \otimes \omega'_k \right\|^2 - \left\| \sum_k u_k \right\|^2,$$

and $\sum_k u_k$ is the off-diagonal $(e, e^\perp)$ block of $\operatorname{Adj}(A)$. So if $\operatorname{Adj}(A) = aI$ then

$$C_2 = \left\| \sum_k f_k \otimes \omega'_k \right\|^2 \geq 0,$$

with equality if and only if $\sum_k f_k \otimes \omega'_k = 0$.

Proof. For a vector u and forms α, γ one has $\iota_w(u \wedge \alpha) = \langle w, u \rangle \alpha - u \wedge \iota_w \alpha$, hence

$$\langle u \wedge \alpha, w \wedge \gamma \rangle = \langle u, w \rangle \langle \alpha, \gamma \rangle - \langle \iota_w \alpha, \iota_u \gamma \rangle. \quad (14)$$

Applying (14) to each cross term with $\alpha = \omega'_l$, $\gamma = \omega'_k$ gives $C_2 = \sum_{k \neq l} [\langle f_k, f_l \rangle \langle \omega'_k, \omega'_l \rangle - \langle u_k, u_l \rangle]$.

Now $f_k = \iota_e \omega_k$ lies in the plane of ω_k , and it is orthogonal to e , so it lies in the image of that plane under the projection to $e^\perp$, which is the plane of ω'_k . Hence $f_k \wedge \omega'_k = 0$, and since $\|f_k\|^2 \|\omega'_k\|^2 = \|\iota_{f_k} \omega'_k\|^2 + \|f_k \wedge \omega'_k\|^2$ for simple ω'_k , this says exactly $\|f_k\|^2 \|\omega'_k\|^2 = \|u_k\|^2$. Therefore

$$\sum_{k \neq l} \langle f_k, f_l \rangle \langle \omega'_k, \omega'_l \rangle = \left\| \sum_k f_k \otimes \omega'_k \right\|^2 - \sum_k \|u_k\|^2, \quad \sum_{k \neq l} \langle u_k, u_l \rangle = \left\| \sum_k u_k \right\|^2 - \sum_k \|u_k\|^2,$$

and subtracting gives the display. For the last claim, polarizing $\langle v, \Theta_k v \rangle = \|\iota_v \omega_k\|^2$ at $v = e$ against $v \in e^\perp$ gives $\langle e, \text{Adj}(A)v \rangle = -\langle v, \sum_k u_k \rangle$, so $\text{Adj}(A) = aI$ forces $\sum_k u_k = 0$. $\square$

Theorem 25 is sharp in the way the data show: over coordinate families $\min_e C_2 = 0$ to machine precision, attained exactly at the coordinate directions, where each summand $f_k \otimes \omega'_k$ vanishes separately, every ω_k either contains e in its plane, so $\omega'_k = 0$, or is orthogonal to it, so $f_k = 0$. That is precisely why the classical recursion has no cross terms. Off that locus the minimum is strictly positive.

The proof reindexes, and doing so gives every level at once.

Theorem 26. *For every $r \geq 2$,*

$$C_r = \sum_{|S|=r-2} \left[Q_S - \|W_S\|^2 \right], \quad \begin{aligned} Q_S &= \sum_{k, l \notin S} \langle f_k, f_l \rangle \langle \omega'_k \wedge \omega'_S, \omega'_l \wedge \omega'_S \rangle, \\ W_S &= \sum_{k \notin S} \iota_{f_k} (\omega'_k \wedge \omega'_S), \end{aligned}$$

and $Q_S = \left\| \sum_{k \notin S} f_k \otimes (\omega'_k \wedge \omega'_S) \right\|^2 \geq 0$.

Proof. Writing $T = S \cup \{k, l\}$ reindexes C_r as $\sum_{|S|=r-2} \sum_{k \neq l \notin S} \langle f_k \wedge \omega'_l \wedge \omega'_S, f_l \wedge \omega'_k \wedge \omega'_S \rangle$. Apply (14) with $\alpha = \omega'_l \wedge \omega'_S$ and $\gamma = \omega'_k \wedge \omega'_S$. The wedge $\omega'_k \wedge \omega'_S$ is a product of vectors, hence simple, and $f_k \wedge (\omega'_k \wedge \omega'_S) = (f_k \wedge \omega'_k) \wedge \omega'_S = 0$ by the proof of Theorem 25; so $\|f_k\|^2 \|\omega'_k \wedge \omega'_S\|^2 = \|\iota_{f_k} (\omega'_k \wedge \omega'_S)\|^2$, and the diagonal terms of the two double sums cancel exactly as before. The Gram matrix of the vectors $\omega'_k \wedge \omega'_S$ is positive semidefinite, so Q_S is the squared norm of $\sum_{k \notin S} f_k \otimes (\omega'_k \wedge \omega'_S)$. $\square$

At $r = 2$ the set S is empty, $W_\emptyset = \sum_k \iota_{f_k} \omega'_k$ vanishes by tightness, and Theorem 26 is Theorem 25. At $r = 3$ it does not vanish, but it simplifies: since $\iota_{f_n} \omega'_n$ lies in the plane of ω'_n , the self term $(\iota_{f_n} \omega'_n) \wedge \omega'_n$ is zero, and using $\sum_k u_k = 0$,

$$W_{\{n\}} = \sum_{k \neq n} \omega'_k \wedge \iota_{f_k} \omega'_n.$$

So the obstruction at $r = 3$ is exactly the failure of the compressed planes to be mutually orthogonal, and it is the only obstruction: everything else is a square.

It is worth being exact about what this settles. Writing the condition $X_e \leq 0$ at $x = 2\sqrt{a}$ in the form

$$\sum_{r \geq 2} (-1)^r C_r (4a)^{2-r} \geq 0, \quad (15)$$

Theorem 25 is the leading coefficient of (15), that is the case $a \rightarrow \infty$. It gives $X_e \leq 0$ outright when $m \leq 5$, where C_2 is the only cross term. That range does not improve on the two-moment bounds above, so Theorem 25 does not by itself enlarge the unconditional range; what it supplies is the first structural reason for the sign.

The obvious next step, bounding C_3 by $4a C_2$ so as to reach $m \leq 7$, is not available, and it fails for a reason worth recording rather than for lack of effort. The ratio $4a C_2 / C_3$ decreases steadily in m , over cubic graphs it runs 3.49 at $m = 8$, 2.00 at $m = 10$ (Petersen), 1.38 at $m = 12$ (Franklin), 1.03 at $m = 14$ (Heawood), and crosses below 1 at $m = 16$, where the Möbius–Kantor graph gives 0.827, then 0.687 (Pappus, $m = 18$), 0.590 (Desargues, $m = 20$) and 0.457 (Nauru, $m = 24$). So $C_3 \leq 4a C_2$ is false in general, and increasingly so.

The reason is visible one level down. For odd r the criterion wants C_r small, that is $\|W_S\|^2$ close to Q_S ; but the ratio $\sum_n \|W_n\|^2 / \sum_n Q_n$ decays, running 0.207, 0.190, 0.154, 0.119, 0.101 over the same graphs at $m = 8, \dots, 16$. So C_3 is essentially $\sum_n Q_n$, an undamped sum of Schur squares, and the contraction W supplies almost no cancellation. Whatever makes (15) hold is therefore the alternation *between* levels, not any cancellation *within* one.

What the same computation shows is that the truncation is what fails, not the conclusion. Writing Σ_R for the partial sum of (15) over $2 \leq r \leq R$:

G	m	Σ_2	Σ_3	Σ_4	Σ_5	Σ_6
cube	8	1.268	0.905	0.926		
Petersen	10	1.624	0.812	0.891	0.891	
Franklin	12	1.713	0.468	0.730	0.713	0.713
Heawood	14	2.007	0.060	0.679	0.603	0.607
Möbius–Kantor	16	2.139	−0.448	0.669	0.458	0.476
Pappus	18	2.142	−0.974	0.739	0.291	
Desargues	20	2.235	−1.552	0.969	0.127	
Nauru	24	2.231	−2.656	1.786		

The values are for one direction per graph, fixed by the seed in `code/hl.Cr.py`, which also regenerates Theorem 26 against the definition of C_r . The terms $|C_r|(4a)^{2-r}$ peak at $r = 3$ and then decay geometrically, at $m = 16$ they run 2.14, 2.59, 1.12, 0.211, 0.0176, so the series converges, but the number of terms needed grows with m : three nearly suffice at $m = 14$, four at $m = 16$, and at $m = 20$ and 24 the partial sums have not settled by Σ_5 . So (15) is not a statement of bounded length, and in particular the two-term reading fails outright from $m = 16$. By Theorem 26 every term is a Schur square minus the square of a contraction, so what a proof must supply is control of the alternating sum of the Q_S across levels.

How much room the inequality actually has is worth measuring, and the answer is that there is a great deal, and that the recursion cannot use any of it. Evaluating the exact ratio

$R_e = F_A/F_{A(e)}$ at $x = 2\sqrt{a}$ on regular graphs, in units of the threshold $x/2 = \sqrt{a}$:

a	graphs	$R_e/(x/2)$	spread
3	cube, Petersen, Franklin, Heawood	1.3662–1.3692	$3.0 \cdot 10^{-3}$
4	$C_8(1, 2)$, $C_{10}(1, 2)$, $C_{12}(1, 2)$	1.3488–1.3497	$8.6 \cdot 10^{-4}$
5	$C_8(1, 2, 4)$, $C_{10}(1, 2, 5)$	1.3246–1.3322	$7.5 \cdot 10^{-3}$

The ratio is a function of a alone to three digits, independent of the graph and of the dimension, and exceeds the threshold by a third. The slack $-X_e/(F_{A(e)}\sqrt{a})$ likewise *increases* with m , running 0.033, 0.044, 0.049, 0.058 over the four cubic graphs above.

That constant can be identified exactly. Averaging the vertex identity over directions gives $\mathbb{E}_e F_{A(e)} = F'_A/m$, which is $F'_A = \sum_i F_{A(e_i)}$ again, so the direction-averaged ratio is

$$\bar{R} = \frac{mF_A}{F'_A} = \left(\frac{1}{m} \sum_i \frac{1}{x - \rho_i} \right)^{-1} = \frac{1}{G(x)},$$

G the Stieltjes transform of the root measure of F_A . For a coordinate family F_A is the matching polynomial, and the Kesten–McKay measure of parameter a has Stieltjes transform $G(z) = 2(a-1)/((a-2)z + a\sqrt{z^2 - 4(a-1)})$. At $z = 2\sqrt{a}$ the radical is *exactly* 2, and everything collapses:

Proposition 27. $\frac{1}{G(2\sqrt{a})} / \sqrt{a} = \frac{\sqrt{a} + 2}{\sqrt{a} + 1} = 1 + \frac{1}{1 + \sqrt{a}}.$

The identity is proved below by the finite and effective route described next; it is not formalized.

The passage from a finite graph to that value is usually made by citing convergence of the matching measure. It need not be: the evaluation point lies strictly outside the support, so the whole step can be made finite and effective, with an explicit rate. We are grateful to Péter Csikvári for pointing us to the mechanism, which is his with P. E. Frenkel [24], after Abért and Hubai for the chromatic polynomial.

Their theorem is that for a monic multiplicative graph polynomial of bounded exponential type, the power sum $p_j(G)$ of the roots is a fixed linear combination of counts of *connected* subgraphs of G ; the matching polynomial is such a polynomial. In Godsil’s formulation [26] the matching-measure moments count closed *tree-like* walks, and the trace of such a walk of length j carries at most $j/2$ edges. Since a subgraph with fewer than $\text{girth}(G)$ edges is a forest, and since in an a -regular graph the number of copies of a fixed tree is $|V(G)|$ times a function of a alone, one gets an identity rather than a limit.

Proposition 28. *Let G be a -regular of girth g . Then $p_j(G)/|V(G)|$ equals the j -th moment of the Kesten–McKay measure of parameter a for every $j < 2g$. The threshold is sharp: for K_4 , K_5 , $K_{3,3}$, $K_{4,4}$, the cube, Petersen and Heawood, the first differing moment is exactly $2g$ whenever one exists at all (`code/moment_girth.py`, exact arithmetic).*

The agreement of moments below $2g$ is proved, by the tree-like walk count of [26]. The sharpness of the threshold is measured on the seven graphs named, not proved.

Because Heilmann–Lieb confines the roots to $[-\rho, \rho]$ with $\rho = 2\sqrt{a-1}$, while the evaluation point is $x = 2\sqrt{a}$, the ratio $\rho/x = \sqrt{1-1/a}$ is strictly below one. A finite geometric expansion with remainder, and no infinite series anywhere, then converts matching moments into matching transforms.

Proposition 29. *Let ν_1, ν_2 be probability measures on $[-\rho, \rho]$ whose moments agree up to order k , and let $x > \rho$. Then*

$$|G_{\nu_1}(x) - G_{\nu_2}(x)| \leq \frac{2\rho^{k+1}}{x^{k+1}(x - \rho)},$$

and if both transforms are at least $c > 0$ the reciprocals differ by at most that quantity divided by c^2 .

Proof. For $|t| \leq \rho < x$ one has the identity $1/(x - t) = \sum_{j \leq k} t^j/x^{j+1} + t^{k+1}/(x^{k+1}(x - t))$, whose right-hand side does not depend on k . Integrating, the polynomial parts agree by hypothesis, and each remainder is at most $\rho^{k+1}/(x^{k+1}(x - \rho))$ in absolute value. The statement for reciprocals is $|1/S_1 - 1/S_2| = |S_2 - S_1|/(S_1 S_2)$. $\square$

Remark 30. Csikvári asks (personal communication, August 2026) whether $\mu_G(2\sqrt{d-1})$ has a combinatorial meaning for d -regular G with $|V(G)|$ even, observing that it should at least be a positive integer. It does, and the answer is in his own paper with Bencs [32]. Their polynomial $R_G(z) = \sum_M (-z)^{|M|} \prod_{v \notin V(M)} (z + d_v - 1)$ is, for d -regular G , a transformation of the matching polynomial: comparing coefficients termwise,

$$R_G(z) = z^{n/2} \mu_G\left(\frac{z + d - 1}{\sqrt{z}}\right).$$

The point $2\sqrt{d-1}$ is not an arbitrary argument of that substitution. Solving $(\sqrt{z} + (d-1)/\sqrt{z}) = 2\sqrt{d-1}$ gives $(\sqrt{z} - \sqrt{d-1})^2 = 0$, so it is exactly the *double root*, attained only at $z = d-1$. Their Corollary 2.7 expands $R_G(w) = \sum_{A \in \mathcal{PF}} 2^{c(A)} (w-1)^{n-|A|}$ over pseudo-forests A , edge sets each of whose components carries at most one cycle, with $c(A)$ the number of cycles. Putting $w = d-1$,

$$\mu_G(2\sqrt{d-1}) = (d-1)^{-n/2} \sum_{A \in \mathcal{PF}(G)} 2^{c(A)} (d-2)^{n-|A|}. \quad (16)$$

At $d = 3$ the weight is 1 and (16) reads: $2^{n/2} \mu_G(2\sqrt{2})$ is the number of pseudo-forests of G counted with multiplicity 2 per cycle. The values are 19, 74, 76, 297, 1162 for K_4 , $K_{3,3}$, the prism, the cube and Petersen.

The transformation, the double root and the evaluation are machine-checked in `RamaLean/PseudoForest.lean`; Corollary 2.7 is [32] and enters as a hypothesis. Identity (16) is verified by brute force in `code/pseudoforest.py` on nine regular graphs of degrees 3 and 4 and orders 4 to 10, in exact arithmetic. The positivity and integrality Csikvári notes in passing are separately machine-checked in `RamaLean/EvenEval.lean`, together with the observation that the parity hypothesis is needed: for $|V(G)|$ odd the value is $2\sqrt{d-1}$ times an integer.

Propositions 28 and 29 together bound the distance from the finite-graph ratio to the value of Proposition 27 by a quantity decaying like $(1 - 1/a)^g$, with no convergence theorem invoked. The bound is not tight, since $|m_j| \leq \rho^j$ discards the $j^{-3/2}$ decay of the Kesten–McKay moments at the edge: at $a = 3$ it gives 0.62, 0.41, 0.28 at girths 4, 5, 6 against observed errors of $3.2 \cdot 10^{-3}$, $9.2 \cdot 10^{-4}$, $2.2 \cdot 10^{-4}$. What it does give, and the citation did not, is an effective statement about a finite graph.

So the cavity ratio exceeds the threshold the induction can propagate by exactly $1/(1 + \sqrt{a})$, and the excess tends to 0 as a grows, which is the analytic shadow of $2\sqrt{a}$ being asymptotically attained.

The limit appears to be attained from above, never crossed, which suggests that Proposition 27 states an infimum rather than merely a limit:

Conjecture 31. *For every weighted 2-plane family with $\text{Adj}(A) = aI$ and every unit e ,*

$$\frac{R_e}{x/2} \geq 1 + \frac{1}{1 + \sqrt{a}} \quad \text{at } x = 2\sqrt{a}.$$

This was not violated in any family tested, over graphs and over general weighted plane families, minimizing over directions; the smallest margin observed was $2.2 \cdot 10^{-4}$, at the highest-girth cubic graph, which is where Proposition 27 says it should be. Conjecture 31 is strictly stronger than $X_e \leq 0$, which is its $c = 1$ case, and by [33] it cannot be proved by this induction either, the recursion propagates $c = 1$ and nothing else, whatever is true. Against the exact values the prediction is 1.366025 at $a = 3$, 1.333333 at $a = 4$ and 1.309017 at $a = 5$, and the error decays with the girth exactly as Kesten–McKay convergence requires: over cubic graphs $3.2 \cdot 10^{-3}$ at girth 4 (the cube), $9.2 \cdot 10^{-4}$ at girth 5 (Petersen), and $2.2 \cdot 10^{-4}$, $1.7 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$ at girth 6 (Heawood, Möbius–Kantor, Pappus, Desargues). None of that slack is available to the argument:

Proposition 32. *Fix $a > 0$ and suppose the inductive hypothesis of Proposition 24 is strengthened from $R \geq x/2$ to $R \geq c(x/2)$ for some $c > 0$. At $x = 2\sqrt{a}$ the recursion sustains this only for $c = 1$.*

Proof. Propagating $R \geq c\sqrt{a}$ requires $c\sqrt{a} \leq 2\sqrt{a} - a/(c\sqrt{a})$, which after clearing the denominator is $a(c - 1)^2 \leq 0$, so $c = 1$. $\square$

There is one hypothesis the argument has not used, and it changes what has to be proved. The class is *not* closed under compression in the tight sense:

Proposition 33. *For a weighted 2-plane family with $\text{Adj}(A) = aI$ and any unit e ,*

$$\text{Adj}(A^{(e)}) = aI_{e^\perp} - F, \quad F = \sum_k f_k f_k^\top, \quad \text{tr } F = a.$$

Proof. For $u, v \perp e$ one has $\iota_u \omega_k = -\langle u, f_k \rangle e + \iota_u \omega'_k$ with $\iota_u \omega'_k \perp e$, so $\langle \iota_u \omega_k, \iota_v \omega_k \rangle = \langle u, f_k \rangle \langle v, f_k \rangle + \langle \iota_u \omega'_k, \iota_v \omega'_k \rangle$; summing and using $\text{Adj}(A) = aI$ gives the display, and $\text{tr } F = \sum_k \theta_k = \langle e, \text{Adj}(A)e \rangle = a$. $\square$

So compressing costs exactly a positive semidefinite matrix of trace a , and the children of a tight family are deficient by a definite amount. Spending that deficiency strengthens the induction. Write $d(e) = a - \langle e, \text{Adj}(A)e \rangle \geq 0$ for the local deficiency, and replace the hypothesis $R \geq x/2$ of Proposition 24 by

$$(H) \quad R_e \geq \frac{x}{2} + \frac{2d(e)}{x}.$$

This is consistent at the base ($m \leq 1$: $R_e = x$, $d = a$, and $x/2 + 2a/x = x$ exactly), and at a tight family $d \equiv 0$, so it still delivers the band.

Proposition 34. *Let $g_k = \langle \hat{f}_k, F \hat{f}_k \rangle$ be the deficiency the children inherit. Then (H) propagates provided*

$$X_e \leq \frac{2F_{A^{(e)}}}{x} \sum_k \frac{\theta_k g_k}{a + g_k},$$

and the estimate is lossless: the two sides of the resulting comparison are identically equal.

Proposition 34 is proved below and is not formalized; the margins quoted for it afterwards are measured.

The right-hand side is *strictly positive*, since $g_k \geq \theta_k > 0$. So the strengthened induction does not require $X_e \leq 0$; it requires X_e below an explicit positive quantity. On the graphs tested (H) held with margin 0.52 to 0.71, and the sufficient condition held with the right-hand side exceeding $|X_e|$ by a factor of five to twenty.

That is worth having because an upper bound on X_e does exist, and the deficiency supplies it. Applying the vertex identity a second time writes each child polynomial as $F_{A_k''} = (F_{A^{(e)}} - q_k)/x$ with $q_k = \langle \hat{f}_k, W^{(e)}(x) \hat{f}_k \rangle$, so $\sum_k \theta_k F_{A_k''} = (aF_{A^{(e)}} - \text{tr}(FW^{(e)}))/x$, and substituting into $X_e = N_e - \sum_k \theta_k F_{A_k''}$ gives a relation between the vertex matrices at two consecutive levels:

Proposition 35. $x X_e = \text{tr}(F W^{(e)}(x)) - x \langle e, W(x) e \rangle - a F_{A^{(e)}}(x)$, and consequently

$$x X_e \leq a \lambda_{\max}(W^{(e)}(x)) - x \langle e, W(x) e \rangle - a F_{A^{(e)}}(x).$$

Proof. The identity is the substitution above. For the bound, $F = \sum_k \theta_k \hat{f}_k \hat{f}_k^T$ is a non-negative combination of unit rank-ones with $\sum_k \theta_k = \text{tr} F = a$, so $\text{tr}(FW^{(e)}) = \sum_k \theta_k q_k \leq (\sum_k \theta_k) \lambda_{\max}(W^{(e)})$; no spectral theory is needed beyond the definition of $\lambda_{\max}$. $\square$

This is the first upper bound on X_e in the development, every earlier statement bounded it below or computed it exactly, and the trace step is not lossy where it matters: on K_5 and K_6 , where $X_e = 0$, it holds with equality. Together with Proposition 34, whose target is strictly positive, the band is therefore reduced to a spectral bound on the child's vertex matrix.

That bound has a clean form in the cavity ratios, and the form explains what is left. Write $r = R_e/(x/2)$ and $r' = (\max_{e'} R'_{e'})/(x/2)$. Since $\lambda_{\max}(W^{(e)}) = F_{A^{(e)}} - x \min_{e'} F_{A''}$ and $\min_{e'} F_{A''} = F_{A^{(e)}}/\max_{e'} R'_{e'}$, the bound $\lambda_{\max}(W^{(e)}) \leq xF_A/a - 3F_{A^{(e)}}$ is exactly

$$r' \leq \frac{1}{2 - r}. \tag{17}$$

Proposition 36. *The Kesten–McKay value of Proposition 27 satisfies (17) automatically: with $u = 1/(1 + \sqrt{a})$ and $r = r' = 1 + u$, one has $(1 + u)(1 - u) = 1 - u^2 \leq 1$, and the slack is exactly*

$$\frac{1}{1 - u} - (1 + u) = \frac{u^2}{1 - u} = \frac{1}{\sqrt{a}(1 + \sqrt{a})}.$$

So the target is not merely consistent with the Kesten–McKay picture, it is *implied* by it, with a slack that vanishes like $1/a$, matching the asymptotic tightness of $2\sqrt{a}$. On the graphs tested (17) holds throughout, and it is tight exactly where $X_e = 0$: at K_5 and K_6 the two sides agree to four digits (1.5673 and 1.5194), while the larger-girth graphs have margin.

What remains unproved is therefore not (17) in the abstract but the gap between the true ratios and their Kesten–McKay values, which is Conjecture 31 together with its analogue one level down.

The constant is therefore not an artefact of the estimate: $\sqrt{a}$ is the fixed point of $\rho \mapsto x - a/\rho$ at $x = 2\sqrt{a}$, the value at which the two roots of $\rho^2 - x\rho + a$ collide. For $x > 2\sqrt{a}$ the inductive hypothesis has an interval of admissible thresholds and the argument has slack; at $x = 2\sqrt{a}$ the interval is a point and it has none. That is exactly why $2\sqrt{a}$, and not the conjectured $2\sqrt{a-1}$, is the ceiling of every argument of this shape. The cross terms $C_r = N_r - \sum_k \theta_k M''_{k,r-1} = \sum_{|T|=r} \sum_{k \neq l \in T} \langle f_k \wedge \omega'_{T \setminus k}, f_l \wedge \omega'_{T \setminus l} \rangle$ were nonnegative in every family tested, and at $r = 2$ that is now Theorem 25. The nonnegativity is not termwise, and it is worth recording that it is not, since the termwise statement is the natural thing to try and it is false: single pairs (k, l) and even whole summands $\sum_{k \neq l \in T}$ take negative values, the smallest observed being $-1.9 \cdot 10^{-3}$ and $-3.8 \cdot 10^{-3}$ over weighted plane families in dimensions six and eight. If $C_r \geq 0$ holds it is therefore a statement about the sum over all T of a given size, not about its terms. Summed, it survives minimization over the sphere, and sharply: for coordinate families $\min_e C_2 = 0$ to machine precision, attained exactly at the coordinate directions, where the classical recursion lives, while for noncommuting projection and general plane families the minimum is strictly positive, growing from 0.048 at $m = 4$ to 0.777 at $m = 8$. Finally $X_e = \sum_{r \geq 2} (-1)^{r-1} C_r x^{m-2r}$ was nonpositive at $x = 2\sqrt{a}$ in every case under direct maximization over the sphere, zero to $4 \cdot 10^{-12}$ for coordinate families, strictly negative otherwise, over graph families in dimensions 4, 5, 6, projection families in dimensions 4, 6 and weighted plane families in dimension 4. The identity of Theorem 23 was checked to 10^{-13} on graph, projection and general weighted plane families at generic directions (`code/hl_Wspec.py`, `code/hl_Xsign.py`).

Both of the bounds that follow are rungs of a single ladder, and it is worth saying what the ladder reaches before giving them. Writing $y = x^2$, the M_r are the elementary symmetric functions of the y -roots, so with p_k the power sums one has $y_{\max}^k \leq p_k$, and any a priori bound $p_k \leq (4a)^k$ gives the band (`MomentLadder.le_of_power_sum.le`). Writing $p_k = (m/2)c_k^k$, the k -th rung covers dimension m exactly when

$$m \leq 2 \left(\frac{4a}{c_k} \right)^k \quad (18)$$

(`reach_of_moment`). Measured on the class, c_k^k is the number of closed walks of length $2k$ from a root of the a -regular tree, so $c_1 = a$ and $c_2 = \sqrt{a(2a-1)}$, which are the two rungs below; that count grows like $(4(a-1))^k k^{-3/2}$, the base being the square of the tree’s spectral radius, against a target of $(4a)^k$. Since $4(a-1) < 4a$, the reach (18) is unbounded in k (`reach_unbounded`): at $a = 3$ it is 146, 509 and 1489 at $k = 5, 7, 9$, against the 8 and 18 of the first two rungs (`code/moments2.py`). What the reach costs is an a priori bound on p_k , and the natural candidate is $p_k \leq (m/2)W_{2k}$ with W_{2k} the tree walk count; for graphs that is Godsil’s path-tree argument, the path tree being a subtree of the universal cover. Off the coordinate case it is false. Weighted families with $\text{Adj}(A) = aI$ exceed it by 1.008, 1.134 and 1.315 at $m = 6, 8, 10$ and $a = 3$, the excess growing in both k and m , so no constant-factor version survives either; coordinate families obey it, with equality at $k = 1, 2$ (`code/momentbound.py`, `code/p16verify.py`). Weakening it to a rate, $p_k \leq (m/2)R^k W_{2k}$, leaves the reach unbounded exactly when $R < a/(a-1)$ (`rate_lt_target`,

`reach_unbounded_of_rate`), and the measured rates sit far inside that: 0.97, 1.02, 1.05, 1.07, 1.09 at $m = 6, \dots, 14$ and $a = 3$ (`code/excessrate.py`). That is still not a route, for a reason that is structural rather than numerical.

Proposition 37. *Let $y_1, \dots, y_n \geq 0$, let $c > 0$, and suppose $\sum_i y_i^k \leq Cc^k$ for every $k \geq 1$, with C independent of k . Then $y_i \leq c$ for every i .*

Proof. If $y_i > c$ then $(y_i/c)^k \leq (\sum_j y_j^k)/c^k \leq C$ for every k , while the left side is unbounded (`MomentLadder.band_of_all_moments`). $\square$

Applied with $c = 4(a-1)R$, a rate bound holding at every rung already forces $y_{\max} \leq 4(a-1)R$. So at $R < a/(a-1)$ the hypothesis is strictly stronger than the band it is meant to prove, and at $R \geq a/(a-1)$ every rung collapses to $m \leq 2$ (`reach_trivial_of_rate`). The ladder has content only at a single finite k , where it buys the finite range (18) and nothing more; the dimension restriction is intrinsic to it.

Two bounds follow unconditionally from real-rootedness alone. Writing $F_A = x^m - M_1x^{m-2} + M_2x^{m-4} - \dots$, one has $M_1 = \frac{1}{2} \text{tr Adj}(A) \leq am/2$ and, using $\|\omega_j \wedge \omega_l\|^2 \geq 1 - \text{tr}(P_j P_l)$, $M_1^2 - 2M_2 \leq \text{tr Adj}(A)^2 - \sum_k c_k^2$. Since the largest root λ satisfies $\lambda^2 \leq M_1$ and $\lambda^4 \leq M_1^2 - 2M_2$, the band $[a - 2\sqrt{a}, a + 2\sqrt{a}]$ holds whenever $m \leq 8$, and whenever $m \leq 16 + (\sum_k c_k^2)/a^2$, which for projection families is $m \leq 32a/(2a-1)$. The second reproduces the known two-moment bound for projections and extends it to all weighted plane families. Adversarial search over rank-two projection families and over random biregular graphs reaches 0.83 of the threshold at $a = 3$ and 0.72 at $a = 4$, with no violation; graphs are marginally more extremal than general projections.

4.6 The reflection for projections

At $b = 2$ the whole question also collapses onto the known half, by a symmetry.

Proposition 38. *Let $P_1, \dots, P_q$ be rank-2 orthogonal projections on $\mathbb{R}^p$ with $\sum_k P_k = aI$. Then $\mu[P_1, \dots, P_q](2a - y) = (-1)^p \mu[P_1, \dots, P_q](y)$; in particular $\lambda_{\min} + \lambda_{\max} = 2a$.*

Proof. The mixed characteristic polynomial is affine-linear in each slot, so it commutes with taking expectations slotwise, and for rank-one slots $\mu[v_1 v_1^\top, \dots] = \det(yI - \sum_k v_k v_k^\top)$, since a multi-affine polynomial is sent by $\prod_k (1 - \partial_k)$ to its value at $z = (-1, \dots, -1)$. Choose an orthonormal pair e_k, f_k spanning the range of P_k and set $v_k = e_k + \varepsilon_k f_k$ with ε_k independent uniform signs. Then $\mathbb{E} v_k v_k^\top = P_k$ and $v_k v_k^\top = P_k + \varepsilon_k S_k$ with $S_k = e_k f_k^\top + f_k e_k^\top$ and $S_k^2 = P_k$, so $\mu(y) = \mathbb{E}_\varepsilon \det((y-a)I - \sum_k \varepsilon_k S_k)$. That is an even or odd function of $t = y - a$ according to the parity of p , because $\varepsilon \mapsto -\varepsilon$ preserves the distribution. $\square$

The last step, which carries the whole proposition, is formalized in `RamaLean/SignReflection.lean` as `sign_avg_reflect`: for any family of matrices $S_1, \dots, S_q$ the sign average $\sum_{\varepsilon \in \{\pm 1\}^q} \det(tI - \sum_k \varepsilon_k S_k)$ satisfies $\rho(-t) = (-1)^n \rho(t)$, with no hypothesis on the S_k ; the shift and the exchange of band edges $2a - (\sqrt{a-1} + 1)^2 = (\sqrt{a-1} - 1)^2$ are `mixedChar_reflect` and `band_edges_swap`. The step identifying the sign average with μ is formalized in the same file: by `sum_prod_sgn` a nonempty sign monomial averages to zero over $\{\pm 1\}^q$, proved by the same involution, so by `sum_multiaffine` the average of a

multi-affine function returns 2^q times its constant term, which for $\det(yI + \sum_k z_k A_k)$ with rank-one A_k is what $\prod_k (1 - \partial_{z_k})$ extracts.

Since $2a - (\sqrt{a-1} + 1)^2 = (\sqrt{a-1} - 1)^2$, the two band edges are exchanged by the reflection, so at $b = 2$ the lower bound is *equivalent* to the upper one. For diagonal families the upper bound is Godsil's path-tree theorem and $b = 2$ is therefore settled, that is Theorem 8 again. For projections it is not: the path tree is a graph construction, and the only bound available in that generality is the Marchenko–Pastur edge $(\sqrt{a} + \sqrt{b})^2$, which is strictly larger than $(s + t)^2$. So at $b = 2$ the projection statement is reduced to its upper half, not closed. The mechanism in the diagonal case is transparent: there $\sum_k \varepsilon_k S_k$ is a signed adjacency matrix, the expectation is Godsil–Gutman, and one recovers $\nu_{S(H)}(y) = \mu_H(y - a)$.

This also locates the obstruction at $b \geq 3$ sharply. Writing $N_k = v_k v_k^\top - P_k$ one has $N_k^2 = (b-2)N_k + (b-1)P_k$, with spectrum $\{b-1, -1^{(b-1)}\}$ on the range of P_k ; the reflection needs this invariant under $N_k \mapsto -N_k$, which happens exactly when $b = 2$. Equivalently: the mean root is always a , since $\sum_i \lambda_i = \text{tr} \sum_k P_k = pa$, while the band centre is $a + b - 2$. The conjecture therefore asserts a skew of exactly $b - 2$, which is the operator form of the displacement in Theorem 16 and is invisible to any reflection or first-moment argument.

The source of that skew can be named exactly. The jump N_k has spectrum $\{b-1, (-1)^{(b-1)}\}$ on the range of P_k , whose moments are

$$m_1 = 0, \quad m_2 = b(b-1), \quad m_3 = b(b-1)(b-2).$$

So the third moment vanishes precisely when $b = 2$: Proposition 38 is not an accident of small b but the vanishing of the jump skewness, and for $b \geq 3$ any argument reaching the lower edge must see a third-order quantity.

5 What this leaves

The biregular case is not a leftover from the refutation; it is where the phenomenon actually lives, and this paper has tried to say how much room it has.

On the positive side the case is genuinely constrained. Complete bipartite graphs never refute it, by a margin $\sqrt{d-1} - h_d/2$ that Gershgorin on the Hermite Jacobi matrix makes positive for every d , and the ratio route closes the inner edge on a nontrivial class. Those are the two proved statements here, and they are not vacuous: the first covers a family whose gap can be made arbitrarily wide, and the second reaches the edge exactly, its certificate degenerating at the gap edge and nowhere earlier.

On the negative side the margin decays like $n^{-2/3}$. The statement is true but tight, and that single fact disqualifies most of what one would try next, including the Gershgorin constant that proves the complete bipartite case: any argument yielding a bound independent of the size is known in advance to be insufficient. What is needed is a Friedman-type edge theorem, in which the roots fill out $\text{spec}(T)$ without escaping it, at the soft-edge scale.

The obstructions are the part we would most want a reader to take away, because each one closes a route that looks open from outside.

- No bound of the form $\text{maxroot } \mu \leq F(a, b, p, q, \text{spec } A_1, \dots, \text{spec } A_q)$ can reduce the problem. Every A_k in a tight rank- b projection family has spectrum $\{1^{(b)}, 0^{(p-b)}\}$, so any such F is constant on the class; the class contains commuting families whose

greatest root tends to the band edge, so F is at best that edge, which is the statement to be proved. Both halves are machine-checked in `RamaLean/SpectralNoGo.lean`: that every functional factoring through the power traces of the blocks agrees on any two families of idempotents with equal block traces (`const_of_factors_through_traces`, from $A^{j+1} = A$ and nothing else), and that an F which is both valid on the class and strong enough to give the target satisfies $F = L$ exactly, so that the bound it yields is the target verbatim (`spectral_bound_is_the_target`). This is why the barrier route yields no reduction, and it applies equally to the Poisson-binomial variance of [33], which is a spectral functional and is constant at $(b/a)(1 - 1/a)$ on the class (`variance_of_idem`). A proof must read a joint invariant of the family. The commutator is one: over random tight projection families the greatest root correlates with $\sum_{i < j} \|[A_i, A_j]\|_F^2$ at -0.33 , -0.50 and -0.73 for C_4 , C_6 and C_8 , and is largest at zero, which is the commuting locus (`code/jointinv.py`). That correlation has since been tested adversarially rather than on random families: hill climbing inside the projection class to *maximise* the greatest root, each block rotated by the exponential of a random antisymmetric generator so that idempotency stays exact, from 76 random starts across C_4 , C_6 , $K_{3,3}$ and the cube at 600 steps each. It accepted 7661 uphill steps and never reached the commuting value, falling short by 0.000031, 0.000857, 0.002763 and 0.010813 respectively, with the commutator at the best point below its starting value in every family (`code/jointinv.py`, full configuration; `--quick` truncates it to two families and 120 steps). It is still not a theorem, and the functional it names is the one a proof would have to read.

The obvious local form of that statement is false, and its failure gives the right one. Both quantities are quadratic forms on the kernel at a commuting family: the greatest root has the curvature form Λ , and since $[A_i, A_j] = \varepsilon([P_i, D_j] + [D_i, P_j]) + O(\varepsilon^2)$ the commutator has $\Gamma(D) = \sum_{i < j} \|[P_i, D_j] + [D_i, P_j]\|_F^2$, a sum of squares. A Lyapunov statement would read $\Lambda \leq -c\Gamma$ with $c > 0$, and that forces $\ker \Lambda \subseteq \ker \Gamma$. It fails: $\ker \Gamma$ is exactly the conjugation orbit, of dimension $\binom{n}{2}$, while $\ker \Lambda$ is strictly larger at $b = 2$, of dimension 8 against 6 at C_4 and 24 against 15 at C_6 , and directions on the cone with $Q = 0$, $\Lambda = 0$ and $\Gamma > 0$ are exhibited at both. So no inequality of that shape holds. What replaces it is visible along those same directions: the root falls at order four where the commutator grows at order two, and the two exponents pair into a constant. At C_4 the ratio $(y_0 - \lambda_{\max})/C^2$ holds at 2.9×10^{-4} to two digits over $t \in [0.02, 0.20]$ while $(y_0 - \lambda_{\max})/t^4$ holds near 9×10^{-7} (`code/lyapunov.py`). The shape to aim at is therefore $\lambda_{\max} \leq y_0 - cC^2$, quadratic in the commutator rather than linear, and the degenerate directions that make the second-order test blind are what distinguishes the two.

Those directions also carry the sharpest local test the extremality claim has been put to, and it passes. On $\ker \Lambda$ the curvature form says nothing, so whether the commuting family is even a local maximum there is decided at order four; a direction with $c_4 \leq 0$ would refute the claim outright. Over 20 directions of $\ker \Lambda$ on the cone at each of C_4 and C_6 , found by solving $Q = 0$ and $\Gamma = 1$ inside $\ker \Lambda$ from random starts, the root falls at every direction and every step size, with $\ker \Lambda$ of dimension 8 against an orbit of 6 at C_4 and 24 against 15 at C_6 . The resolvable fourth-order constants run from 7.7×10^{-5} to 1.73 at C_4 and from 2.8×10^{-2} to 3.16 at C_6 ; three directions at C_4 are reported as unresolved rather than quoted, their $y_0 - \lambda_{\max}$ falling to about

4×10^{-12} , which is the precision floor of the root solve. A direction does not determine a curve, order two leaving the off-diagonal blocks of the correction free, so the sign is also checked along curves whose second-order term is perturbed inside the kernel: over 240 curves at each family, twenty directions carrying twelve perturbed corrections apiece, the smallest constant found is 1.2×10^{-6} at C_4 and 2.3×10^{-2} at C_6 , both positive (`code/quartic.py`). Since Γ is normalised to 1 on these directions the same constants are c_4/Γ^2 , which is the constant in the quadratic shape. The shape is local of necessity and not by caution: $y_0 - \lambda_{\max}$ is bounded on the class while C^2 is not, so the ratio falls as the commutator grows and no global constant can work.

- A barrier argument that sees only the trace and $\sum_k P_k = aI$ is reading the Marchenko–Pastur law, and $A_k = (b/p)I_p$ satisfies both while exceeding the tree band already at $p = 4$. No such argument can reach either edge.
- Compressions cannot reach the inner end at all: a compression takes values the original excludes, so no argument phrased in Rayleigh quotients can exclude them.
- The coefficient ladder’s reach is unbounded in the moment order, but a bound holding at every rung with a constant free of the order already gives the conclusion (`MomentLadder.band_of_all_moments`). The dimension restriction on the two unconditional bounds is therefore intrinsic to the method, not an artefact of stopping early, and the natural a priori input — the tree walk count — is false off the coordinate case.

One question the paper does close, rather than obstruct, is the value of the constant. On the coordinate locus the mixed characteristic polynomial of a tight rank- b family is the matching polynomial of an (a, b) -biregular incidence graph in the squared variable ([33]), so Xu’s inequality is a theorem there at every b and, by [33], the constant $(\sqrt{a-1} + \sqrt{b-1})^2$ is attained and cannot be lowered. That removes a possibility the numerics had left open — that the constant is honest at $b = 2$ and slack at $b \geq 3$, the adversarial search having stalled at 0.739 and 0.726 there — and it fixes what any proof of the general statement has to deliver. It also locates the difficulty: the bridge fails off the coordinate locus, and that is where the whole content of the conjecture now sits.

What survives all three obstructions is the ratio recursion, which bounds ratios of matching polynomials rather than Rayleigh quotients, and that is where we would put further effort. The remaining gap is stated plainly: the certificate closes for six of the seven families tested, and the defect in the seventh is a right vertex whose child ratio sits well below the interval a uniform certificate can express. Closing that is the concrete open problem this paper ends on.

6 Related work

Two ingredients are classical and are used, not claimed. The factorization of the characteristic polynomial of a cyclic cover into twisted determinants at roots of unity is the block-circulant diagonalization [20, 21]; in the infinite-cover limit it is the Floquet–Bloch decomposition of a periodic discrete operator, and $y = -P/(2Q)$ is the corresponding Floquet discriminant. The identity $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ is [7]. Closest in form is McSorley and Feinsilver [19]. For a path whose edges are cyclically labelled with t labels they establish a (τ, Δ) -recurrence

$\Phi_N = \tau\Phi_{N-1} - \Delta\Phi_{N-2}$, which is the algebraic skeleton of (2), and their Theorem 4.4 solves it as $\Delta^{N/2}U_N(\tau/2\sqrt{\Delta})$, exactly the form used here. In the univariate specialization their τ is the matching polynomial of the cycle C_t and $\Delta = (-1)^t x_1 \cdots x_t$ is a monomial; passing between their counting normalization and the characteristic one used here is the homogenization $V_d \mapsto c^d V_d$, under which a monomial Δ becomes $\mu_{G-V(C)} = 1$. Their Theorems 4.4 and 4.5 together give $U_{dn+n-1} = U_d(T_n)U_{n-1}$, the identity invoked in Section 1 to pass between Theorem 3 and Hall’s conjecture, so that bridge was already available in 2009. Their parameter is the number of label periods along a path and their graphs are paths, cycles and rooted trees; there is no covering and no cycle with trees attached. What is added here is the pair $(\mu_G, \mu_{G-V(C)})$, whose second entry is a genuine polynomial recording the attached trees rather than a monomial, together with the covering interpretation that makes the recurrence a statement about $\mu_{d,G}$.

7 The formalization

Everything asserted here is machine-checked in Lean 4/Mathlib unless labelled otherwise; the sources are `RamaLean/` in the accompanying repository and carry no `sorry` and no custom axioms. The standing blocker is that Mathlib has no matching polynomial and no mixed characteristic polynomial, so the identities connecting the combinatorial and operator forms are stated rather than derived. Statements are labelled `VERIFIED`, `HEURISTIC` or `CONJECTURE` in the sources, with their plain meaning: a `HEURISTIC` claim is supported by computation and is not proved. The file-by-file record is in [33].

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